SGA 4, Exposé VIII

Fiber Functors, Supports, Cohomological Study of Finite Morphisms

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1. Topological invariance of the étale topos

Theorem 1.1. Let $f : X' \to X$ be an *integral, surjective, radicial* morphism. Then the base-change functor

$$f^* : X_{\text{ét}} \longrightarrow X'_{\text{ét}}$$

is an equivalence of sites, that is, an equivalence of categories which is continuous, as is every quasi-inverse functor. Consequently the functors

$$f_* : \widetilde{X'_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}},$$
$$f^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

are quasi-inverse equivalences of categories.

The first assertion is well known when f is of finite presentation or when f is flat (SGA 1, IX, 4.10 and 4.11); the general case reduces to the finite-presentation case. Indeed, it is already known that the functor f^* is fully faithful (SGA 1, IX, 3.4). It remains to prove that f^* is essentially surjective, i.e. that every Z' étale over X' "comes from" a Z étale over X. Using the fact that f is a universal homeomorphism, together with the full faithfulness of f^*, one is reduced to the case where X, X', and Z' are affine, say the spectra of rings A, A', and B'. Writing A' as the filtered inductive limit of its finite-type A-subalgebras A'_i, the algebra B' comes from an étale algebra B'_i over some A'_i (cf. EGA IV, 8). Thus one is reduced to the case where A' is integral and of *finite type* over A, hence finite over A. Then A' is A-isomorphic to A''/J, where $A'' \simeq A[t_1, \dots, t_n]$ and J is an ideal of A''. Writing J as the inductive limit of its finite-type subideals J_i, and putting $A'_i = A''/J_i$, one again has $A' = \varinjlim_i A'_i$, so that B' comes from an étale algebra B'_i over some A'_i (loc. cit.). On the other hand, since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is surjective, the same is true of $\operatorname{Spec} A'_i \to \operatorname{Spec} A$; furthermore one checks easily that, since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is radicial, the same is true of $\operatorname{Spec} A'_i \to \operatorname{Spec} A$ for i sufficiently large. For example, apply EGA IV, 1.9.9 to $S = \operatorname{Spec} A$: if $S_i \subset S$ denotes the set of points s for which the fibre of $X_i = \operatorname{Spec} A'_i$ at s has separable rank one, then the S_i are constructible subsets of S (EGA IV, 9.7.9), form an increasing family, and their union is S, by the hypothesis on $S' = \operatorname{Spec} A' \to S$ and the fact that the fibres of the S'_i are noetherian schemes. Hence $S = S_i$ for i sufficiently large. This reduces us to the case where A' is one of the A'_i, hence of finite presentation over A.

This proves the first assertion of Theorem 1.1. The fact that f_* and f^* are quasi-inverse equivalences follows immediately.

Corollary 1.2. Let F be an abelian sheaf on X, and let F' be its inverse image on X'. Then the canonical homomorphisms

$$H^i(X, F) \longrightarrow H^i(X', F') \tag{*}$$

are isomorphisms. Likewise, if F' is an abelian sheaf on X' and F is its direct image on X, the canonical homomorphisms $(*)$ are isomorphisms.

Examples 1.3. Here are some frequent applications of Theorem 1.1.

a) X' is a subscheme of X with the same underlying set as X, i.e. it is defined by a nil-ideal I.

b) X is a scheme over a separably closed field k, k' is an algebraic closure of k, and $X' = X \otimes_k k'$.

c) Let X be a geometrically unibranch scheme, for example an algebraic curve over a field k having only "cuspidal" singularities, excluding in particular ordinary double points. If X' is the normalization of X_{red}, then by definition $X' \to X$ is radicial, so that Theorem 1.1 applies.

Remarks 1.4. A morphism f as in Theorem 1.1 is a *universal homeomorphism*, i.e. it is a homeomorphism and remains one after every base extension. Conversely, if f is a universal homeomorphism, then f satisfies the hypotheses of Theorem 1.1 (cf. EGA IV, 8.11.6 when f is of finite presentation, and EGA IV, 18.12.11 in general). This explains the title of the present section.

Concerning Example 1.3(c), note explicitly that if X is an algebraic curve, say over an algebraically closed field k, having at least one singular point which is not unibranch, for instance an ordinary double point, and if X' denotes the normalization of X_{red}, then the conclusion of Corollary 1.2 already fails for H^1 with constant coefficients; compare SGA 1, I, 11(a), and SGA 1, IX, 5.

2. Sheaves on the spectrum of a field

Proposition 2.1. Let k be a field, let $\bar{k}$ be a separable closure of k, let $\pi = \mathrm{Gal}(\bar{k}/k)$ be its topological Galois group, and put $X = \operatorname{Spec} k$, $\bar{X} = \operatorname{Spec} \bar{k}$, on which π acts on the left. Consider the canonical functor

$$i : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}}$$

(defined by $i(X')(X'') = \mathrm{Hom}_X(X', X'')$), and the canonical functor

$$j : \widetilde{X_{\text{ét}}} \longrightarrow \mathcal{B}_\pi$$

from the category of sheaves on $X_{\text{ét}}$ to the category $\mathcal{B}_\pi$ of discrete sets endowed with a continuous left action of π (cf. IV, 2.7), defined by

$$j(F) = \varinjlim_{\alpha} F(\operatorname{Spec} k_\alpha),$$

where k_α runs through the finite subextensions of $\bar{k}$. Then i and j are equivalences of categories. Consequently the topos $\widetilde{X_{\text{ét}}}$ is equivalent (IV, 3.4) to the classifying topos $\mathcal{B}_{\pi^\bullet}$.

The composite functor

$$ji : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}} \longrightarrow (\pi\text{-}\mathbf{Set})$$

is an equivalence of categories by the fundamental theorem of Galois theory, in the form of SGA 1, V. Since $\mathcal{B}_\pi$ is plainly a topos (II, 4.14), the same is true of $X_{\text{ét}}$. Moreover a family $(X_i \to X)$ of morphisms in $X_{\text{ét}}$ is covering if and only if it is surjective, i.e. if and only if its image in $\mathcal{B}_\pi$ is surjective, or equivalently covering for the canonical topology of $\mathcal{B}_\pi$. This shows that the topology of $X_{\text{ét}}$ is indeed its canonical topology. Hence i is an equivalence, and therefore so is j.

Corollary 2.2. The functor j of Proposition 2.1 induces an equivalence between the category of abelian sheaves on $X = \operatorname{Spec} k$ and the category of Galois π-modules.

Indeed, the latter are precisely the abelian-group objects of the topos $\mathcal{B}_\pi$. Similarly, by transport of structure one obtains the following.

Corollary 2.3. Let F be an abelian sheaf on $X = \operatorname{Spec} k$, and let $M = j(F)$ be the associated Galois π-module. Then there is a canonical isomorphism of δ-functors in F:

$$H^*(X, F) \simeq H^*(\pi, M),$$

where the right hand side denotes Galois cohomology, studied for example in Serre's course *Cohomologie galoisienne*.

Corollary 2.4. Assume that k is separably closed. Then the functor

$$\Gamma : \widetilde{X_{\text{ét}}} \longrightarrow (\mathbf{Set})$$

is an equivalence of categories. If F is an abelian sheaf on $X = \operatorname{Spec} k$, then $H^i(X, F) = 0$ for $i \neq 0$. In other words, the topos $\widetilde{X_{\text{ét}}}$ is a punctual topos (IV, 2.2).

Corollary 2.5. Let k' be a separably closed extension of a separably closed field k, put $X = \operatorname{Spec} k$ and $X' = \operatorname{Spec} k'$, and let $u : X' \to X$ be the canonical morphism. Then the functor

$$u^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

is an equivalence of categories.

3. Fiber functors associated with geometric points of a scheme

3.1

Let X be a scheme. We shall call a *geometric point* of X any X-scheme ξ which is the spectrum of a separably closed field Ω. Giving such a point is therefore equivalent to giving an ordinary point $x \in X$ and a separably closed extension Ω of the residue field $k(x)$. The most frequent case for us will be that where Ω is a separable closure of $k(x)$, which we shall generally denote by $k(\bar{x})$, and the corresponding geometric point of X by $\bar{x}$. For fixed $x \in X$, the fields $k(\bar{x})$, respectively the geometric points $\bar{x}$, are determined only up to non-unique $k(x)$-isomorphism, respectively non-unique X-isomorphism.

Remarks 3.2. In most geometric questions it is more natural to restrict to algebraically closed Ω. The different convention used here is special to the study of the étale topology. It is the convention adopted in the definition of the fundamental group in SGA 1, V, 7. Note however that the developments of loc. cit. are equally valid with the convention adopted here, since the property of Ω used there is that every étale covering of $\operatorname{Spec} \Omega$ is trivial, precisely the assertion that Ω is separably closed.

Definition 3.3. Let X be a scheme, let ξ be a geometric point of X, and let $u : \xi \to X$ be its structural morphism. The *geometric fiber functor* associated with ξ, denoted $F \mapsto F_\xi$, is the composite functor

$$\widetilde{X_{\text{ét}}} \xrightarrow{u^*} \widetilde{\xi_{\text{ét}}} \xrightarrow{\Gamma_\xi} (\mathbf{Set}).$$

Equivalently, by Corollary 2.4, a geometric point ξ of X defines a point of the topos $\widetilde{X_{\text{ét}}}$ (IV, 6.1), and one takes the associated fiber functor.

3.4

Since the functor Γ_ξ of sections over ξ is an equivalence of categories (2.4), knowing the fiber functors $F \mapsto F_\xi$ is essentially equivalent to knowing the inverse-image functors u^*. Moreover it follows immediately from Corollary 2.5 that if ξ is a geometric point of X and there exists an X-morphism $\xi' \to \xi$, i.e. if ξ' corresponds to a separably closed extension Ω' of Ω, then the canonical homomorphism $F_\xi \to F_{\xi'}$ is a functorial isomorphism

$$F_\xi \simeq F_{\xi'}.$$

Thus, for the study of fiber functors, one may, after replacing Ω by the separable algebraic closure of $k(x)$ in Ω, restrict to the case where $k(\xi)$ is a separable closure $\overline{k(x)}$ of $k(x)$. The fiber functors corresponding to geometric points of X localized at the same point $x \in X$ are isomorphic, non-canonically.

In accordance with the conventions of 3.1, we shall denote by $F_{\bar{x}}$ a fiber functor corresponding to a chosen separable closure $\overline{k(x)}$ of $k(x)$.

We also record an evident transitivity property, which explains the technical usefulness of not restricting exclusively to geometric points defined by separable closures of residue fields. Let $f : X' \to X$ be a morphism of schemes, and let $u' : \xi' \to X'$ be a geometric point of X'. Then $u = fu' : \xi' \to X$ is a geometric point of X, which we denote by ξ. For these geometric points one has a functorial isomorphism in $F \in \mathrm{Ob}\,\widetilde{X_{\text{ét}}}$:

$$f^*(F)_{\xi'} \simeq F_\xi.$$

This follows from the transitivity formula for inverse-image functors,

$$(fu)^* \simeq u^* f^*.$$

Theorem 3.5. Let X be a scheme.

a) For every geometric point ξ of X, the fiber functor $F \mapsto F_\xi$, from $\widetilde{X_{\text{ét}}}$ to (**Set**), commutes with arbitrary inductive limits, indexed by categories in $\mathcal{U}$, and with finite projective limits. In particular it sends sheaves of groups, respectively abelian sheaves, and so on, to groups, respectively abelian groups, and so on.

b) As x runs over the points of X, the fiber functors $F_{\bar{x}}$ form a conservative family: a morphism $u : F \to G$ of sheaves on X is an isomorphism if and only if, for every $x \in X$, the corresponding morphism $u_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$ is an isomorphism.

Taking account of the exactness properties in (a), we immediately record the following formal consequences of (b).

Corollary 3.6. Let $u : F \to G$ be a morphism of sheaves. Then u is a monomorphism, respectively an epimorphism, if and only if for every $x \in X$, the map $u_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$ is injective, respectively surjective.

Corollary 3.7. Let $u, v : F \to G$ be two morphisms of sheaves on X. Then $u = v$ if and only if, for every $x \in X$, one has $u_{\bar{x}} = v_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$. In particular, if u, v are two sections of F, then $u = v$ if and only if $u_{\bar{x}} = v_{\bar{x}}$ in $F_{\bar{x}}$ for every $x \in X$.

Corollary 3.8. Let $F \to G \to H$ be a sequence of two morphisms of sheaves on X. The sequence is exact if and only if, for every $x \in X$, the corresponding sequence $F_{\bar{x}} \to G_{\bar{x}} \to H_{\bar{x}}$ is exact.

We leave to the reader the deduction of these corollaries (cf. IV, 6.5.2), although they are also proved in part in the proof below.

Assertion (a) follows from the exactness properties already noted in Exposé VII, 1.4 for every inverse-image functor, such as $u^* : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$, and from the fact that $\widetilde{\xi_{\text{ét}}} \to (\mathbf{Set})$ is an equivalence. To prove (b), note the following formula, which gives a way to *compute* fiber functors.

Proposition 3.9. Let $\xi \xrightarrow{u} X$ be a geometric point of X, and let C_ξ be the category of "ξ-pointed étale X-schemes", i.e. pairs (X', u'), where $X' \in \operatorname{Ob} X_{\text{ét}}$ and $u' : \xi \to X'$ is an X-morphism. Then the opposite category C_ξ° is filtering, and for a variable sheaf F, respectively presheaf P, on X, there is a canonical functorial isomorphism

$$F_\xi \simeq \varinjlim_{C_\xi^\circ} F(X'), \qquad (aP)_\xi \simeq \varinjlim_{C_\xi^\circ} P(X'),$$

where aP denotes the sheaf associated with P.

Indeed, first note that for every presheaf P on $\xi_{\text{ét}}$, if aP is its associated sheaf, the natural homomorphism

$$P(\xi) \longrightarrow aP(\xi)$$

is an isomorphism; this follows immediately, for example, from the explicit computation of aP as $L(LP)$ (II, 3.19). Let $\varphi : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ be the inverse-image functor. Then $u^* = \varphi^*$ is, by III, 2.3(d), the composite $a\varphi^\bullet i$, where $i : \widetilde{X_{\text{ét}}} \to \widehat{X_{\text{ét}}}$ is the inclusion functor and $a : \widehat{\xi_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ is the associated-sheaf functor. By the preceding remark,

$$u^*(F)(\xi) = (\varphi^\bullet i(F))(\xi).$$

Thus it remains to apply the explicit expression for $\varphi^\bullet$ given in the proof of the corresponding general formula. The analogous assertion for a presheaf P is proved in the same way, using the functorial isomorphism $\varphi^* a(P) \simeq a\varphi^*(P)$. The fact that C_ξ is filtering, which remains true for *any* X-scheme ξ, not necessarily a geometric point, follows immediately from the existence of finite projective limits in C_ξ; this in turn follows from the fact that $X_{\text{ét}}$ is a subcategory of $(\mathrm{Sch})_{/X}$ stable under finite projective limits.

Now let $u : F \to G$ be a morphism of sheaves on X such that, for every $x \in X$, the map $u_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$ is a monomorphism. We prove that u is a monomorphism. It suffices to show that if $X' \in \operatorname{Ob} X_{\text{ét}}$ and $\varphi, \psi \in F(X')$ have the same image under u, then $\varphi = \psi$. Replacing X by X' and using the transitivity of fiber functors, we may suppose $X = X'$. For every $x \in X$, we have $u(\varphi)_{\bar{x}} = u(\psi)_{\bar{x}}$, hence $u_{\bar{x}}(\varphi_{\bar{x}}) = u_{\bar{x}}(\psi_{\bar{x}})$, and therefore $\varphi_{\bar{x}} = \psi_{\bar{x}}$. By Proposition 3.9, there exists $X'_x \in \operatorname{Ob} X_{\text{ét}}$, whose image contains x, such that φ and ψ have the same inverse image on X'_x. As x varies over X, the X'_x form a covering family of X; hence $\varphi = \psi$.

This proves that if all $u_{\bar{x}}$ are monomorphisms, then so is u. Suppose in addition that all $u_{\bar{x}}$ are epimorphisms. We prove that u is an epimorphism, hence an isomorphism. It remains to show that for every $X' \in \operatorname{Ob} X_{\text{ét}}$ and every $\psi \in G(X')$, there exists $\varphi \in F(X')$ with $u(X')(\varphi) = \psi$. Again we may suppose $X' = X$. By Proposition 3.9, for every $x \in X$ there exists $X'_x \in \operatorname{Ob} X_{\text{ét}}$, with image containing x, and an element $\varphi_x \in F(X'_x)$ whose image in $G(X'_x)$ is the inverse image of ψ. Since u is a monomorphism, the φ_x agree on the fibre products $X'_x \times_X X'_y$, and hence come from a well-determined element $\varphi \in F(X)$. Then $u(\varphi) = \psi$, since the inverse images on the X'_x agree. $\square$

Scholium 3.10. Theorem 3.5 implies that the collection of fiber functors for the étale topology has the same essential properties as in the usual theory of sheaves on a topological space. We shall use Theorem 3.5 and its corollaries very frequently, and for this reason we shall usually refrain from referring to them explicitly.

3.11

For every $x \in X$ we shall also, by abuse of language, denote by x the X-scheme $\operatorname{Spec} k(x)$, with its usual structural morphism

$$i_x : x = \operatorname{Spec} k(x) \longrightarrow X.$$

It gives rise to a canonical functor

$$i_x^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{x_{\text{ét}}}.$$

If F is a sheaf on X, put $F_x = i_x^*(F)$. Thus F_x is a sheaf on $x = \operatorname{Spec} k(x)$, and as such can also be identified with an étale scheme over x, by Proposition 2.1. It depends functorially on F, and the functor $i_x^* : F \mapsto F_x$ again commutes with arbitrary inductive limits and finite projective limits, by Exposé VII, 1.4.

If $\bar{x} = \operatorname{Spec} \overline{k(x)}$, the fiber functor $F \mapsto F_{\bar{x}}$ is canonically isomorphic to the composite of $F \mapsto F_x$ with the functor j of Proposition 2.1, taking into account that the latter is isomorphic to the fiber functor associated with the geometric point $\operatorname{Spec} \bar{k}$ of $\operatorname{Spec} k$. It follows immediately from this and Theorem 3.5 that the family of functors $(F \mapsto F_x)$, for $x \in X$, is still conservative. If $k(x)$ is separably closed, i.e. $x = \bar{x}$, then the notations F_x and $F_{\bar{x}}$ are strictly contradictory, the first denoting an object of $\widetilde{x_{\text{ét}}}$, the second a set; but by Proposition 2.1 this is an inessential contradiction.

3.12

The preceding remarks show that, if $\bar{x} = \operatorname{Spec} \overline{k(x)}$, then for every sheaf F on X the group $\pi_x = \operatorname{Gal}(\overline{k(x)}/k(x))$ acts naturally on the left on the fiber set $F_{\bar{x}}$. Thus $F \mapsto F_{\bar{x}}$ may in fact be regarded as a functor

$$\widetilde{X_{\text{ét}}} \longrightarrow \mathcal{B}_{\pi_x},$$

whose knowledge is essentially equivalent, by Proposition 2.1, to that of the functor $F \mapsto F_x$.

This shows, in an important special case, that contrary to what happens for sheaves on topological spaces, fiber functors generally have nontrivial automorphisms. In general, Theorem 7.9 below will determine all homomorphisms from one fiber functor to another.

Remark 3.13. The following is a sometimes useful generalization of Theorem 3.5(b). Let E be a subset of X such that, for every étale morphism $f : X' \to X$, the inverse image $E' = f^{-1}(E)$ is very dense in X' (EGA IV, 10.1.3), i.e. every open subset U of X' containing E' is equal to X'. Then the fiber functors $F_{\bar{x}}$ on $\widetilde{X_{\text{ét}}}$, for $x \in E$, already form a conservative family. Consequently Corollaries 3.6, 3.7, and 3.8 remain valid when one restricts to $x \in E$. This is seen immediately by repeating the proof of Theorem 3.5(b). The preceding result applies in particular in the following situations.

a) X is a Jacobson scheme (EGA IV, 10.4.1), for example locally of finite type over a field or over $\operatorname{Spec} \mathbb{Z}$, and E is the set of closed points of X.

b) X is locally of finite type over a scheme S, and E is the set of points of X which are closed in their fibre.

c) X is locally noetherian, and E is the set of $x \in X$ such that $\overline{\{x\}}$ is a finite set, i.e. a semi-local scheme of dimension ≤ 1; compare EGA IV, 10.5.3 and 10.5.5.

4. Strictly local rings and schemes

4.1

Recall that, following Azumaya, a local ring A is called *henselian* if it satisfies the following equivalent conditions.

(i) Every finite A-algebra B is a direct product $\prod_i B_i$ of local rings. The B_i are then necessarily isomorphic to the localizations $B_{\mathfrak{m}_i}$, where the $\mathfrak{m}_i$ run through the maximal ideals of B.

(ii) Every A-algebra B which is the localization of a finite-type A-algebra C at a prime ideal $\mathfrak{p}$ over the maximal ideal $\mathfrak{m}$ of A, and such that $B/\mathfrak{m}B$ is finite over $k = A/\mathfrak{m}$, is finite over A.

(iii) As in (ii), but assuming C to be étale over A at $\mathfrak{p}$, and $k \to B/\mathfrak{m}B$ to be an isomorphism.

(iv) Hensel's lemma, in its classical form, holds for A: for every monic polynomial $F \in A[T]$, denoting by $\dot{F} \in k[T]$ its reduction, and for every factorization

$$\dot{F} = \dot{F}_1 \dot{F}_2$$

of $\dot{F}$ as a product of two coprime monic polynomials, $\dot{F}_1$ and $\dot{F}_2$ lift, necessarily uniquely, to $F_1, F_2 \in A[T]$ such that $F = F_1 F_2$.

For the proof of these equivalences and the general properties of henselian rings, see EGA IV, 18.5.11. Recall that if A is any local ring, one proves, following Nagata, that there is a local homomorphism

$$A \longrightarrow A^h$$

with A^h henselian, universal for local homomorphisms from A to henselian local rings. The ring A^h is called the henselization of A. The construction in EGA IV, 18.6, different from Nagata's, consists in putting

$$A^h = \varinjlim_i A_i,$$

where A_i runs through the A-algebras essentially of finite type and essentially étale over A, i.e. localizations of an étale A-algebra, such that $A \to A_i$ is local and the residue extension $k(A_i)/k(A)$ is trivial. This construction shows in particular that A^h is flat over A, and that if A is noetherian then so is its henselian closure.

Definition 4.2. A local ring A is called *strictly local* if it satisfies one of the following equivalent conditions.

(i) A is henselian and its residue field is separably closed.

(ii) Every local homomorphism $A \to B$, where B is the localization of an étale A-algebra, is an isomorphism.

A scheme is called a *strictly local scheme* if it is isomorphic to the spectrum of a strictly local ring.

The equivalence of (i) and (ii) follows immediately from 4.1. In geometric form, 4.2(ii), respectively 4.1(iii), says that for every scheme X étale over $Y = \operatorname{Spec} A$, and every point x of the fibre X_y, respectively every point of X_y rational over $k(y)$, where y is the closed point of Y, there exists a section of X over Y passing through y, necessarily unique.

Definition 4.3. Let X be a scheme, let $\mathcal{O}_{X_{\text{ét}}}$ be the sheaf on $X_{\text{ét}}$ defined by

$$\mathcal{O}_{X_{\text{ét}}}(X') = \Gamma(X', \mathcal{O}_{X'}),$$

compare Exposé VII, 2(c), and let $u : \xi = \operatorname{Spec} \Omega \to X$ be a geometric point of X. The *strictly local ring of* X *at* ξ, denoted $\mathcal{O}_{X,\xi}$, or simply $\mathcal{O}_\xi$, is the fibre of the sheaf $\mathcal{O}_{X_{\text{ét}}}$ at the point ξ. Its spectrum is called the *strict localization of* X *at* ξ.

By Proposition 3.9, the fibre $\mathcal{O}_{X,\xi}$ has the description

$$\mathcal{O}_{X,\xi} = \varinjlim \Gamma(X', \mathcal{O}_{X'}), \tag{*}$$

where X' runs through the filtering opposite category of étale X-schemes pointed by ξ, i.e. endowed with an X-morphism $\xi \to X'$. By transitivity of inductive limits, one may replace $\Gamma(X', \mathcal{O}_{X'})$ on the right by $\mathcal{O}_{X',x'}$, where x' is the image of ξ in X', and obtains

$$\mathcal{O}_{X,\xi} = \varinjlim_i A_i, \tag{**}$$

where A_i runs through the algebras essentially of finite type and étale over $A = \mathcal{O}_{X,x}$, endowed with a k-homomorphism $k(A_i) \to \Omega$. Here x denotes the image of ξ in X. The transition homomorphisms among the A_i are the local homomorphisms of A-algebras $A_i \to A_j$ making the corresponding triangle commute,

$$\begin{tikzcd} k(A_i) \arrow[r] \arrow[dr] & k(A_j) \arrow[d] \\ & \Omega, \end{tikzcd}$$

which implies that if there exists an admissible homomorphism from A_i to A_j, it is unique. Thus the inductive limit $(**)$ is a limit over an increasing filtered ordered set.

4.4

The description $(**)$ shows that the strictly local ring of X at ξ depends essentially only on the usual local ring $A = \mathcal{O}_{X,x}$, where $x = u(\xi)$, and on the extension Ω of $k = k(A)$. It is also called the *strict henselization* of A, relative to the chosen separably closed extension Ω of k, and is denoted A^{hs}. We refer to EGA IV, 18.8 for the general properties of A^{hs}. We merely point out that the construction above again shows that A^{hs} is flat over A, and noetherian if A is noetherian. On the other hand, one easily proves (loc. cit.) that the local homomorphism

$$A \longrightarrow A^{hs}$$

endowed with the k-homomorphism

$$k(A^{hs}) \longrightarrow \Omega$$

solves the universal problem, relative to the data A and the extension Ω of k, of finding local homomorphisms

$$A \longrightarrow A'$$

with A' strictly local, endowed with a k-homomorphism

$$k(A') \longrightarrow \Omega.$$

In particular, A^{hs} *is* a strictly local ring, justifying the terminology in Definition 4.3.

4.5

Keep the notation of Definition 4.3 and choose an affine open neighbourhood U of x. In Proposition 3.9 one may plainly restrict to those X' which are affine over U. These then form a pseudo-filtered projective system of affine schemes, and we are in the general situation of Exposé VII, 5 (cf. VI, 5.12(b)). By the expression $(*)$ for $\mathcal{O}_{X,\xi}$, one has an X-isomorphism

$$\varprojlim_{\substack{X' \text{ étale} \\ \text{affine over } U \\ \xi\text{-pointed}}} X' \simeq \operatorname{Spec} \mathcal{O}_{X,\xi}, \tag{***}$$

which makes precise the intuitive geometric meaning of $\operatorname{Spec} \mathcal{O}_{X,\xi}$ as the *limit of the étale neighbourhoods of* ξ.

Proposition 4.6. Let X be a strictly local scheme, and let x be its closed point, hence a geometric point of X. Then the functor $F \mapsto \Gamma(X, F) : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$ is canonically isomorphic to the fiber functor $F \mapsto F_x$, and therefore commutes with arbitrary inductive limits, not necessarily filtered.

Indeed, by Definition 4.2(ii), the final object X of the category of x-pointed étale X-schemes considered in Proposition 3.9 dominates all the other objects. Hence $\varinjlim F(X')$ is canonically isomorphic to $F(X)$. □

In particular, the induced functor Γ on the category of abelian sheaves is exact. Therefore:

Corollary 4.7. Under the hypotheses of Proposition 4.6, for every abelian sheaf F on X,

$$H^q(X, F) = 0 \qquad \text{for } q \neq 0.$$

Corollary 4.8. Let X be a scheme, let ξ be a geometric point of X, let $\bar{X} = \operatorname{Spec} \mathcal{O}_{X,\xi}$ be the corresponding strictly localized scheme, let F be a variable sheaf on X, and let $\bar{F}$ be its inverse image on $\bar{X}$. Then there is a functorial isomorphism

$$F_\xi \simeq \Gamma(\bar{X}, \bar{F}).$$

Indeed, ξ may also be regarded as a geometric point of $\bar{X}$, and it suffices to apply Proposition 4.6 and the transitivity of fibres (3.4).

5. Application to computing the fibres of $R^q f_*$

5.1

Let

$$f : X \to Y$$

be a morphism of schemes, and let F be an abelian sheaf on X. We want to determine $R^q f_*(F)$. By Theorem 3.5, this amounts, up to minor qualifications, to knowing the geometric fibres $R^q f_*(F)_{\bar{y}}$, for $y \in Y$. Now by Exposé V, 5.1, $R^q f_*(F)$ is the sheaf associated with the presheaf

$$\mathcal{H}^q : Y' \longmapsto H^q(X \times_Y Y', F)$$

on Y. Therefore, by Proposition 3.9,

$$R^q f_*(F)_{\bar{y}} \simeq \varinjlim H^q(X \times_Y Y', F), \tag{5.1.1}$$

where the inductive limit is taken over the filtering opposite category of the Y' étale over Y pointed by $\bar{y}$. Choosing an affine open neighbourhood U of y, one may restrict in the limit to the cofinal category of the Y' which are affine and lie over U.

Introduce

$$\bar{Y} = \mathrm{Spec}(\mathcal{O}_{Y,\bar{y}}) = \varprojlim Y'$$

(cf. 4.5), and

$$\bar{X} = X \times_Y \bar{Y}.$$

In the projective system of the $X \times_Y Y'$, obtained from that of the Y' by base change along $X \to Y$, the transition morphisms are affine. Thus one is under the general hypotheses of Exposé VII, 5.1, and by the construction of $\varprojlim$ in loc. cit. there is a canonical isomorphism

$$\bar{X} = \varprojlim (X \times_Y Y').$$

Let $\bar{F}$ be the inverse image of F on $\bar{X}$. One obtains a canonical homomorphism

$$\varinjlim_{Y'} H^q(X \times_Y Y', F) \longrightarrow H^q(\bar{X}, \bar{F}),$$

and hence, by comparison with (5.1.1), a canonical homomorphism, functorial in F,

$$R^q f_*(F)_{\bar{y}} \longrightarrow H^q(\bar{X}, \bar{F}). \tag{5.1.2}$$

Assume now that $f : X \to Y$ is quasi-compact and quasi-separated. Then the same is true of $X \times_Y Y' \to Y'$, and since Y' is affine the schemes $X \times_Y Y'$ are quasi-compact and quasi-separated. Using (5.1.1) and the limit-passage theorem of Exposé VII, 5.8, we obtain the following.

Theorem 5.2. Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes, let F be an abelian sheaf on X, let y be a point of Y, let $\bar{y}$ be the geometric point above y associated with a separable closure $k(\bar{y})$ of $k(y)$, let $\bar{Y} = \mathrm{Spec}\, \mathcal{O}_{Y,\bar{y}}$ be the corresponding strict localization, put $\bar{X} = X \times_Y \bar{Y}$, and let $\bar{F}$ be the inverse image of F on $\bar{X}$. Then the canonical homomorphism (5.1.2) is an isomorphism:

$$R^q f_*(F)_{\bar{y}} \simeq H^q(\bar{X}, \bar{F}).$$

This statement practically reduces the determination of the fibres of a sheaf $R^q f_*(F)$ to the determination of the cohomology of a prescheme over a strictly local scheme, and will be used constantly below. Technically, Theorem 5.2 explains the important role of henselian rings and strictly local rings in the study of étale cohomology.

Remark 5.3. The statement remains valid for $R^0 f_*(F) = f_*(F)$, with F an arbitrary sheaf of sets. Also, Theorem 5.2 remains valid for $R^1 f_*(F)$ when F is a sheaf of groups, not necessarily commutative, using the usual definition of $R^1 f_*(F)$ (cf. Giraud's thesis).

5.4

Assume now that f is a *finite* morphism. Then $\bar{f} : \bar{X} \to \bar{Y}$ is finite, and since $\bar{Y}$ is strictly local it follows that $\bar{X}$ is a finite sum of strictly local schemes $\bar{X}_i$. Therefore, using Corollary 4.7,

$$H^q(\bar{X}, \bar{F}) = 0 \qquad \text{for } q \neq 0. \tag{5.4.1}$$

On the other hand, the components $\bar{X}_i$ of $\bar{X}$ correspond to the points $\bar{x}_i$ of $\bar{X}$ above the closed point $\bar{y}$ of $\bar{Y}$, i.e. to the points of

$$\bar{X}_{\bar{y}} = X_y \otimes_{k(y)} k(\bar{y}).$$

These points may also be regarded as geometric points of X, and $\bar{X}_i$ is then nothing other than the strict localization of X at $\bar{x}_i$. Thus, by Corollary 4.8,

$$H^0(\bar{X}_i, \bar{F}) \simeq F_{\bar{x}_i},$$

and hence

$$H^0(\bar{X}, \bar{F}) = \prod_i F_{\bar{x}_i}, \tag{5.4.2}$$

a functorial isomorphism in the sheaf of sets F; here it is not necessary to restrict to abelian sheaves. Taking Theorem 5.2 and Remark 5.3 into account, formulas (5.4.1) and (5.4.2) give the following.

Proposition 5.5. Let $f : X \to Y$ be a finite morphism of preschemes, and let y be a point of Y. For every sheaf F on X, there is a functorial isomorphism

$$f_*(F)_{\bar{y}} \simeq \prod_{\bar{x} \in X_y \otimes_{k(y)} k(\bar{y})} F_{\bar{x}}.$$

Consequently formation of $f_*(F)$ commutes with every base change $y' \to Y$. If F is an abelian sheaf, then

$$R^q f_*(F) = 0 \qquad \text{for } q > 0.$$

The first formula in Proposition 5.5 is in fact independent of Theorem 5.2 and of the general limit-passage theorem of Exposé VII, 5.7. It implies, by Theorem 3.5, that $F \mapsto f_*(F)$ is an exact functor on the category of abelian sheaves, again giving $R^q f_*(F) = 0$ for $q \neq 0$. Here is a slight variant of Proposition 5.5.

Corollary 5.6. Let $f : X \to Y$ be an integral morphism. Then for every abelian sheaf F on X,

$$R^q f_*(F) = 0 \qquad \text{for } q \neq 0.$$

Moreover, the functor f_* on sheaves of sets commutes with every base change $Y' \to Y$.

Indeed, by Theorem 5.2 we are reduced to proving that, when Y is strictly local, $H^q(X, F) = 0$ for every abelian sheaf F on X. Write $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, with B an integral A-algebra. Writing $B = \varinjlim B_i$, where the B_i run through the finite-type subalgebras of B, which are even finite over A, we have $X = \varprojlim_i X_i$, where $X_i = \operatorname{Spec} B_i$. By Exposé VII, 5.13, it is enough to prove that $R^q f_*(F_i) = 0$ for $q \neq 0$ with F_i an abelian sheaf on X_i, and this follows from Proposition 5.5. The final assertion of Corollary 5.6 is proved by the same reduction to Proposition 5.5.

Corollary 5.7. Let $f : X \to Y$ be an immersion. Then for every sheaf F on X, the canonical morphism

$$f^* f_*(F) \longrightarrow F$$

is an isomorphism.

Since f factors as the product of an open immersion and a closed immersion, and Corollary 5.7 is trivial for an open immersion (IV, No. 3), one is reduced to the case where f is a closed immersion. It is enough to prove that for every $x \in X$, the corresponding homomorphism on geometric fibres

$$(f^* f_*(F))_{\bar{x}} \longrightarrow F_{\bar{x}}$$

is bijective. By transitivity of fibres (3.4), the left hand side is just the fibre $f_*(F)_{\bar{x}}$, so it remains to check that the canonical homomorphism

$$f_*(F)_{\bar{x}} \longrightarrow F_{\bar{x}}$$

is bijective. This follows immediately from Proposition 5.5.

Remark 5.8. Using Proposition 5.5 and proceeding as in Corollary 5.6, one easily proves that if $f : X \to Y$ is integral and F is a sheaf of groups on X, not necessarily commutative, then $R^1 f_*(F)$ is the final sheaf on Y; compare Remark 5.3.

6. Supports

Let U be a Zariski open subset of the scheme X. Then $U \in \mathrm{Ob}\, X_{\text{ét}}$, and in fact U is a subobject of the final object X of $X_{\text{ét}}$. Hence it defines a subobject $\widetilde{U}$ of the final object of $\widetilde{X_{\text{ét}}}$, i.e. an open subtopos of the étale topos of X (IV, 8.3).

Proposition 6.1. The preceding map $U \mapsto \widetilde{U}$ is an isomorphism from the ordered set of open subsets of X, in the Zariski sense, onto the set of open subobjects of the étale topos $\widetilde{X_{\text{ét}}}$.

Since $X_{\text{ét}} \to \widetilde{X_{\text{ét}}}$ is fully faithful, the map $U \mapsto \widetilde{U}$ plainly preserves order, i.e. $(U \subset V) \Leftrightarrow (\widetilde{U} \subset \widetilde{V})$, so in particular it is injective. To prove surjectivity, let F be a subsheaf of the final sheaf $\widetilde{X}$, and consider the objects of $X_{\text{ét}}/F$, i.e. the étale schemes X' over X such that $F(X') \neq \varnothing$. Since $X' \to X$ is étale, it is an open map (EGA IV, 2.4.6); in particular its image $\mathrm{Im}(X')$ is open. Let U be the union of the opens $\mathrm{Im}(X')$ for $X' \in \mathrm{Ob}\, X_{\text{ét}}/F$. Since the family $X' \to U$, for $X' \in \mathrm{Ob}\, X_{\text{ét}}/F$, is surjective, hence covering, one concludes that $U \in \mathrm{Ob}\, X_{\text{ét}}/F$, hence $X_{\text{ét}}/F = X_{\text{ét}}/U$, and therefore $F = U$. □

Taking Proposition 6.1 into account, we may therefore speak without ambiguity of an open of X, without specifying whether we mean this in the usual Zariski sense or in the sense of the étale topology.

Corollary 6.2. Let U be an open subset of X, and let $j : U \to X$ be the canonical immersion. Then the functor

$$j^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{U_{\text{ét}}}$$

induces an equivalence of categories

$$\widetilde{X_{\text{ét}}}/\widetilde{U} \longrightarrow \widetilde{U_{\text{ét}}}.$$

Indeed, for every site C in which finite projective limits exist, and for every subobject U of the final object e of C, the functor $j : C \to C_{/U}$ defined by $j(S) = S \times U$ is a morphism of sites and $j^* : \widetilde{C} \to \widetilde{C/U}$ induces an equivalence $\widetilde{C}/\widetilde{U} \to \widetilde{C/U}$. It remains only to combine this general fact with the canonical isomorphism $U_{\text{ét}} \simeq X_{\text{ét}}/U$.

Pictorially, Corollary 6.2 says that the operations of restricting to an open, in the usual sense for schemes on the one hand and in the sense of topoi on the other, are compatible. The following is the analogous compatibility for restriction to a closed subset.

Theorem 6.3. Let X be a scheme, let Y be a closed subscheme of X, let $U = X - Y$ with its induced structure, and let $i : Y \to X$ and $j : U \to X$ be the canonical immersions. Then the functor

$$i_* : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

is fully faithful. Moreover, for $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$, the sheaf F is isomorphic to one of the form $i_*(G)$ if and only if $j^*(F)$ is isomorphic to the final sheaf on U.

Proof. Since i_* and i^* are adjoint, the full faithfulness of i_* is equivalent to the assertion that the functorial homomorphism

$$i^*(i_*(G)) \longrightarrow G$$

is an isomorphism; this is just Corollary 5.7. On the other hand, if $G \in \operatorname{Ob} \widetilde{Y_{\text{ét}}}$, then by Corollary 6.2 one checks trivially that $j^*(i_*(G))$ is the final sheaf on U.

Conversely, suppose $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$ is such that $j^*(F)$ is the final sheaf. We prove that F is isomorphic to a sheaf of the form $i_*(G)$, or equivalently that the canonical homomorphism

$$F \longrightarrow i_* i^* F$$

is an isomorphism. It is enough to check this on all geometric fibres. If $x \in U$, this is precisely the hypothesis on F. If $x \in Y$, transitivity of fibres reduces us to checking that the homomorphism on fibres induced by

$$i^*(F) \longrightarrow i^*(i_* i^*(F))$$

is an isomorphism. But this homomorphism is an isomorphism by Corollary 5.7, applied to i and to i^*F. This completes the proof.

Corollary 6.4. The functor i_* induces an equivalence between the category of abelian sheaves on Y and the category of abelian sheaves on X whose restriction to $U = X - Y$ is zero.

In accordance with common usage, we shall therefore often identify an abelian sheaf on Y with an abelian sheaf on X which is zero on U.

6.5

By Proposition 6.1 we know how to interpret the opens of the topos $\mathcal{E} = \widetilde{X_{\text{ét}}}$ as ordinary opens U of X. By Theorem 6.3, with this identification, the residual topos $\mathcal{E}_{\complement U}$ of IV, 3 is canonically equivalent to $\widetilde{Y_{\text{ét}}}$, where $Y = X - U$ is endowed with any induced scheme structure. We may therefore apply to this situation the results of IV, 3, especially the description of sheaves F on X in terms of triples (F', F'', u), where F' is a sheaf on Y, F'' a sheaf on U, and u a homomorphism from F' to $i^* j_*(F'')$. We shall freely use below the notation $i^!$, $j_!$ of IV, 3, denoting functors

$$j_! : (\widetilde{U_{\text{ét}}})_{\text{ab}} \longrightarrow (\widetilde{X_{\text{ét}}})_{\text{ab}},$$
$$i^! : (\widetilde{X_{\text{ét}}})_{\text{ab}} \longrightarrow (\widetilde{Y_{\text{ét}}})_{\text{ab}},$$

where the subscript ab denotes the category of abelian sheaves. These functors give rise to the two exact sequences of IV, 3.7.

6.6

In view of the preceding developments and of the general terminology introduced in IV, 8.5, it is natural to introduce, for an abelian sheaf F on X, or a section φ of such a sheaf, the notion of the *support* of F, respectively of φ, as the closed subset, in the usual Zariski sense, complementary to the largest open over which the object in question vanishes.

In the present situation it is also appropriate to replace the general notation of Exposé V, 4.3, $H^q(\mathcal{E}_{\complement U}, F)$, $\mathcal{H}^q_{\complement U}(F)$, by the notation $H^q_Y(X, F)$, $\mathcal{H}^q_Y(F)$. The first denotes an abelian group, and the second an abelian sheaf on Y, or equivalently an abelian sheaf on X which is zero on U. Thus $\mathcal{H}^q_Y = R^q i^!$, and so forth. We refer to Exposé V, 6 for the general properties of these functors.

7. Specialization morphisms of fiber functors

7.1

We saw in No. 4 how to associate to every geometric point ξ of the scheme X a strictly local X-scheme
$$\bar{X}(\xi) = \operatorname{Spec} \mathcal{O}_{X,\xi},$$
which in fact depends only on the geometric point above the image x of ξ defined by the separable closure $\bar{k(x)}$ of $k(x)$ in $\Omega = k(\xi)$. We shall often restrict below to geometric points $\xi = \operatorname{Spec} \Omega$ which are *separable algebraic* over X, i.e. such that Ω is a separable closure of $k(x)$, in other words $\xi = \bar{x}$. Call an X-scheme Z a *strict localization of* X if it is X-isomorphic to a scheme of the form $\operatorname{Spec} \mathcal{O}_{X,\bar{x}}$. Then
$$\xi \longmapsto \operatorname{Spec} \mathcal{O}_{X,\xi} = \bar{X}(\xi)$$
is an equivalence from the category of separable algebraic geometric points over X, with only isomorphisms as morphisms, to the category of X-schemes which are strict localizations of X, again with only isomorphisms as morphisms.

The latter assertion is no longer true if one allows all X-morphisms, because in the second category there may be X-morphisms which are not isomorphisms. We therefore make the following definition.

Definition 7.2. Let ξ and ξ' be two geometric points of the scheme X. A *specialization arrow* from ξ' to ξ is an X-morphism between the corresponding strict localizations
$$\bar{X}(\xi') \longrightarrow \bar{X}(\xi).$$
We say that ξ is a specialization of ξ', or that ξ' is a generization of ξ, if there exists a specialization arrow from ξ' to ξ.

Specialization arrows compose in the evident way. Thus, taking specialization arrows as morphisms, the geometric points of X form a category, equivalent, as is the full subcategory of separable algebraic geometric points over X, to the full subcategory of $(\mathrm{Sch})_{/X}$ consisting of the strict localizations of X.

Lemma 7.3. Let X be a scheme, let Z be an X-scheme isomorphic to a pseudo-filtered projective limit of étale X-schemes X_i with affine transition morphisms (Exposé VII, 5.1), let ξ' be a geometric point of X, and let $Z' = \bar{X}(\xi')$ be the corresponding strict localization. Then:

a) the restriction map
$$\operatorname{Hom}_X(Z', Z) \longrightarrow \operatorname{Hom}_X(\xi', Z)$$
is bijective;

b) the two sides are nonempty if and only if the image x' of ξ' in X lies in the image of Z;

c) if T is a Z-scheme, then T is a strict localization of Z if and only if it is a strict localization of X.

Proof. For (a), one immediately reduces to the case where Z is one of the X_i, i.e. where Z is étale over X; after the base change $Z' \to X$ one may suppose $Z' \simeq X$, and the conclusion follows from Definition 4.2(ii). For (b), note that if x' is the image of a point z of Z, then $k(z)$ is necessarily a separable algebraic extension of $k(x')$, hence admits a $k(x')$-homomorphism into $k(\xi')$. Assertion (c) is an easy exercise left to the reader.

Definitions 7.2 and Lemma 7.3 imply the following.

Proposition 7.4. Let ξ and ξ' be two geometric points of the scheme X. Restriction defines a bijection from the set $\mathrm{Hom}_X(\bar{X}(\xi'), \bar{X}(\xi))$ of specialization arrows from ξ' to ξ, to the set of X-morphisms from ξ' to $\bar{X}(\xi)$.

Corollary 7.5. The geometric point ξ is a specialization of ξ' if and only if the same is true for their images x, x' in X, i.e. if and only if x belongs to the closure of $\{x'\}$.

Since $\mathrm{Spec}\, \mathcal{O}_{X,\xi} \to \mathrm{Spec}\, \mathcal{O}_{X,x}$ is faithfully flat (4.4), it is surjective, and one only has to apply the second assertion of Lemma 7.3.

Corollary 7.6. For every scheme X, let $\mathcal{P}t(X)$ be the category of geometric points over X, or equivalently of separable algebraic geometric points over X, with specialization arrows as morphisms. Let ξ be a geometric point of a scheme X, and let $\bar{X}(\xi)$ be the corresponding strict localization. Then there is an equivalence of categories

$$\mathcal{P}t(\bar{X}(\xi)) \xrightarrow{\sim} \mathcal{P}t(X)_{/\xi},$$

which associates to every geometric point ξ' of $\bar{X}(\xi)$ the corresponding geometric point over X, together with the specialization morphism to ξ deduced from the structural morphism $\xi' \to \bar{X}(\xi)$ by Proposition 7.4.

In other words, giving a generization ξ' of the geometric point ξ is essentially equivalent to giving a geometric point of $\bar{X}(\xi)$. By Lemma 7.3(c), the corresponding homomorphism $\bar{X}(\xi') \to \bar{X}(\xi)$ makes $\bar{X}(\xi')$ the strict localization of $\bar{X}(\xi)$ relative to ξ'.

7.7

For every geometric point ξ of X and every sheaf F on X, interpret the fibre F_ξ as $\Gamma(\bar{X}(\xi), \bar{F}(\xi))$, where $\bar{F}(\xi)$ is the inverse image of F on $\bar{X}(\xi)$ (Corollary 4.8). Then every specialization arrow

$$u : \xi' \longrightarrow \xi$$

induces a homomorphism, functorial in F,

$$u^* : F_\xi \longrightarrow F_{\xi'},$$

called the *specialization homomorphism* associated with u. It is clear from the transitivity of inverse images of sheaves that for a composite specialization arrow one has

$$(wu)^* = u^* w^*.$$

7.8

Recall that if $\mathcal{E}$ is a topos, a *fiber functor*, or by abuse of language a point of $\mathcal{E}$, is a morphism from the final topos (**Set**) to $\mathcal{E}$, i.e. a functor

$$\varphi : \mathcal{E} \longrightarrow (\mathbf{Set})$$

which commutes with arbitrary inductive limits and finite projective limits. The set of fiber functors of $\mathcal{E}$ is regarded as the set of objects of a full subcategory of $\mathrm{Hom}(\mathcal{E}, (\mathbf{Set}))$, called the *category of fiber functors* of the topos $\mathcal{E}$; its opposite is called the *category of points of* $\mathcal{E}$, denoted $\mathcal{P}t(\mathcal{E})$, cf. IV, 6.1.

When $\mathcal{E} = \widetilde{X_{\text{ét}}}$, where X is a scheme, we defined in 3.3 and 7.7 a functor

$$\mathcal{P}t(X) \longrightarrow \mathcal{P}t(\widetilde{X_{\text{ét}}}). \tag{*}$$

Here the left hand side is defined in Corollary 7.6. With this notation we have:

Theorem 7.9. Let X be a scheme. The functor $(*)$ is an equivalence from the category of geometric points over X, with specialization arrows as morphisms, to the category of points of the étale topos $\widetilde{X_{\text{ét}}}$, the latter being the opposite of the category of fiber functors on $\widetilde{X_{\text{ét}}}$.

Since this theorem will not be used later in the seminar, we give only a sketch of proof, in which we take some liberties with questions of universes.

a) *The functor in question is fully faithful.* For every $X \in \mathrm{Ob}\, X_{\text{ét}}$, let $X^{\vee} : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$ be defined by

$$X^{\vee}(F) = \mathrm{Hom}(X, F) = F(X).$$

Note that the fiber functor

$$\mathcal{E}_{\xi} : F \longmapsto F_{\xi}$$

may be written as

$$\mathcal{E}_{\xi} \simeq \varinjlim_{i} X_i^{\vee},$$

where the X_i are affine schemes étale over X, indexed by a certain filtering category (cf. 4.3 and 4.5). Thus for every functor $\varphi : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$,

$$\mathrm{Hom}(\mathcal{E}_{\xi}, \varphi) \simeq \varprojlim_{i} \mathrm{Hom}(X_i^{\vee}, \varphi).$$

On the other hand, one has the well-known bifunctorial isomorphism

$$\mathrm{Hom}(X_i^{\vee}, \varphi) \simeq \varphi(X_i).$$

When φ is of the form $\mathcal{E}_{\xi'}$, hence

$$\varphi \simeq \varinjlim_{j} (X'_j)^{\vee},$$

there is a natural bijection

$$\mathrm{Hom}(\mathcal{E}_{\xi}, \mathcal{E}_{\xi'}) \simeq \varprojlim_{i} \varinjlim_{j} \mathrm{Hom}_X(X'_j, X_i). \tag{*}$$

On the other hand, in the category of schemes,

$$\bar{X}(\xi) = \varprojlim_{i} X_i, \qquad \bar{X}(\xi') = \varprojlim_{j} X'_j,$$

so that

$$\mathrm{Hom}_X(\bar{X}(\xi'), \bar{X}(\xi)) \simeq \varprojlim_{i} \mathrm{Hom}_X(\bar{X}(\xi'), X_i).$$

Since X_i is locally of finite presentation over X,

$$\mathrm{Hom}_X(\bar{X}(\xi'), X_i) \simeq \varinjlim_j \mathrm{Hom}_X(X'_j, X_i),$$

and therefore

$$\mathrm{Hom}_X(\bar{X}(\xi'), \bar{X}(\xi)) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_X(X'_j, X_i). \tag{**}$$

Comparing $(*)$ and $(**)$ gives the desired conclusion, up to a compatibility check left to the reader.

b) *The functor in question is essentially surjective.*

7.9.1

First observe that if C is a site in which finite projective limits are representable, and $\widetilde{C} = \mathcal{E}$ is the corresponding topos, then every fiber functor φ on $\mathcal{E}$ can be represented as a filtered inductive limit of functors of the form

$$\check{Y}(F) = F(Y) = \mathrm{Hom}(\widetilde{Y}, F), \qquad Y \in \mathrm{Ob}\, C,$$

where $\widetilde{Y}$ is the sheaf associated with Y, as follows. Consider the category $C_{/\varphi}$ of pairs (Y, ξ), with $Y \in \mathrm{Ob}\, C$ and $\xi \in \varphi(\widetilde{Y})$, morphisms being defined in the evident way. There is an evident homomorphism

$$\varinjlim_{(Y,\xi) \in C^{\circ}_{/\varphi}} \check{Y} \longrightarrow \varphi. \tag{*}$$

Indeed,

$$\mathrm{Hom}\left(\varinjlim_{C_{/\varphi}} \check{Y}, \varphi\right) \simeq \varprojlim_{C_{/\varphi}} \mathrm{Hom}(\check{Y}, \varphi) \simeq \varprojlim_{C_{/\varphi}} \varphi(\widetilde{Y}).$$

There is a canonical element of $\varprojlim_{C_{/\varphi}} \varphi(\widetilde{Y})$, assigning to each (Y, ξ) the element $\xi \in \varphi(Y)$. Since finite projective limits exist in C, and the functor $Y \mapsto \varphi(\widetilde{Y})$ commutes with them, finite projective limits exist in $C_{/\varphi}$. A fortiori $C^{\circ}_{/\varphi}$ is filtering. Using the fact that every sheaf F is an inductive limit of sheaves of the form $\widetilde{Y}$, and that φ commutes with inductive limits, one easily concludes that $(*)$ is bijective, giving the asserted representation.

7.9.2

Also note that if $\mathcal{E}$ is a topos and Y an object of $\mathcal{E}$, giving a fiber functor ψ for $\mathcal{E}_{/Y}$ is equivalent to giving a pair (φ, ξ), where φ is a fiber functor for $\mathcal{E}$ and $\xi \in \varphi(Y)$. To $\psi : \mathcal{E}_{/Y} \to (\mathbf{Set})$ one associates (φ, ξ), where φ is the composite $Z \mapsto \varphi(Z) = \psi(Z \times Y)$, and where $\xi \in \varphi(Y) = \psi(Y \times Y)$ is the image of the unique element of $\psi(Y)$ by the diagonal morphism of $Y \times Y$.

7.9.3

Return now to the site $C = X_{\text{ét}}$, and let $\varphi : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$ be a fiber functor on the étale topos of X. Restricting φ to the full subcategory of $\widetilde{X_{\text{ét}}}$ formed by the opens of this topos, or equivalently by Proposition 6.1 by the opens U of X, gives a map

$$\varphi_0 : \mathcal{U}(X) \longrightarrow \{0, 1\}$$

commuting with arbitrary suprema and finite infima. Here $\varphi(U)$ is either empty or reduced to one point, and $\varphi_0(U)$ is defined to be 0 or 1 accordingly.

Using the fact that every irreducible closed subset of X has a unique generic point, one sees easily that φ_0 is defined by a unique point $x \in X$, by the condition

$$\varphi_0(U) = 1 \quad \text{if and only if} \quad x \in U,$$

i.e.

$$\varphi(U) = \varnothing \quad \Longleftrightarrow \quad x \notin U.$$

Let $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$. The image of F in the final sheaf X is an open U, and since

$$\varphi(F) \longrightarrow \varphi(U)$$

is surjective, $\varphi(F) \neq \varnothing$ if and only if $\varphi(U) \neq \varnothing$, i.e. if and only if $x \in U$. Let $C_{/\varphi}$ be the category of pairs (X', ξ), where $X' \in \operatorname{Ob} X_{\text{ét}}$ and $\xi \in \varphi(X')$. We noted in 7.9.1 that $C^{\circ}_{/\varphi}$ is connected and filtering. Moreover, by 7.9.2 and the preceding discussion applied to X' in place of X, to an object (X', ξ) there is associated a unique point $x' \in X'$ such that, for an étale morphism $X'' \to X'$, ξ lies in the image of $\varphi(X'') \to \varphi(X')$ if and only if x' lies in the image of X''. It follows that the scheme $\varprojlim X'$, taken over the category C of the (X', ξ), exists and is a strict localization Z of X, corresponding to a geometric point ξ and to a fiber functor $\mathcal{E}_\xi$. For every sheaf F,

$$\mathcal{E}_\xi(F) \simeq \varinjlim_{C_\varphi} F(X').$$

By 7.9.1, φ is isomorphic to $\mathcal{E}_\xi$, completing the proof of Theorem 7.9.

8. Two spectral sequences for integral morphisms

Proposition 8.1. Let $f : X \to Y$ be an integral surjective morphism, and let F be an abelian sheaf on Y. For every integer i, let $\mathcal{H}^i(F)$ be the presheaf on $(\mathrm{Sch})_{/Y}$ defined by

$$\mathcal{H}^i(F)(Z) = H^i(Z, F_Z),$$

where F_Z is the inverse image of F on Z. Then there is a spectral sequence, functorial in F,

$$H^*(X, F) \Longleftarrow E_2^{pq} = H^p(X/Y, \mathcal{H}^q(F)),$$

the *descent spectral sequence.*

Of course, for a presheaf G on $(\mathrm{Sch})_{/Y}$, the symbol $H^p(X/Y, G)$ denotes the p-th relative Cech cohomology group, defined by the complex $C^n(X/Y, G) = G(X^{n+1})$, where X^m denotes the m-fold power in $(\mathrm{Sch})_{/Y}$. To prove Proposition 8.1, let $p_n : X^{n+1} \to Y$ be the projection, and put

$$A^n = p_{n*}(\mathbb{Z}_{X^{n+1}}),$$

where $\mathbb{Z}_{X^{n+1}}$ denotes the constant sheaf $\mathbb{Z}$ on X^{n+1}. The A^n are the components of a simplicial abelian sheaf on Y, hence of a complex A^* of abelian sheaves on Y. There is an evident homomorphism

$$\mathbb{Z}_Y \longrightarrow A^0. \tag{*}$$

Lemma 8.2. The complex A^*, endowed with the homomorphism $(*)$, is a resolution of $\mathbb{Z}_Y$. More generally, for every abelian sheaf F on Y, the complex $A^* \otimes_{\mathbb{Z}} F$ is a resolution of F.

Proof. We may assume that Y is affine, say $Y = \operatorname{Spec} A$, and $X = \operatorname{Spec} B$, with B an integral A-algebra. Write $B = \varinjlim_i B_i$, where the B_i run over the finite-type, hence finite, A-subalgebras of B. Then $X = \varprojlim X_i$, with $X_i = \operatorname{Spec} B_i$. Using Exposé VII, 5.11, the augmented complex $A^*(X/Y)$ is the inductive limit of the augmented complexes $A^*(X_i/Y)$. This reduces us to the case where f is finite. It is enough to prove that for every geometric point $\bar{y}$ of Y, the complex $A^*_{\bar{y}}$ is a resolution of $\mathbb{Z}_{\bar{y}}$, and remains so after tensoring with F. If $X_{\bar{y}}$ is the fibre of X at $\bar{y}$, Proposition 5.5 shows that $A^*_{\bar{y}}$ is just the analogous complex $A^*(X_{\bar{y}}/\bar{y})$. This reduces us to the case where Y is the spectrum of a separably closed field k. Using Examples 1.3(a) and (b), we may even assume k algebraically closed and X reduced; thus X is of the form I_Y, where I is a finite set. Then $A^* \otimes F$ identifies with the trivial cochain complex of the indexing set I, with coefficients in $F_{\bar{y}}$, and is indeed a resolution of $F_{\bar{y}}$.

As a consequence of Lemma 8.2 we obtain a spectral sequence

$$H^*(Y, F) \Longleftarrow E_2^{pq} = H^p(H^q(Y, A^* \otimes F)),$$

and it remains to identify the initial term. There is a canonical homomorphism

$$A^n \otimes F \longrightarrow p_{n*}p_n^*(F),$$

which is an isomorphism, again by reduction to the finite case and passage to fibres. Using Proposition 5.5, it follows that

$$H^q(Y, A^n \otimes F) \simeq H^q(X^{n+1}, p_n^* F) = \mathcal{H}^q(F)(X^{n+1}),$$

giving the announced initial term in Proposition 8.1.

Remark 8.3.

a) If Y is covered by a locally finite family of closed subsets Y_i, and if $X = \coprod_i Y_i$, then the canonical morphism $f : X \to Y$ is finite, and the initial term of the spectral sequence in Proposition 8.1 is described as the cohomology of the complex defined by the groups $H^q(Y_{i_0 \dots i_p}, F|Y_{i_0 \dots i_p})$, where $Y_{i_0 \dots i_p} = Y_{i_0} \cap \dots \cap Y_{i_p}$. This is the analogue of Leray's spectral sequence for a locally finite closed covering of an ordinary topological space (TF, Chapter II, 5.2.4). The latter can also be generalized to a spectral sequence relative to a "finite" morphism, i.e. a proper morphism with finite fibres, by proceeding as in Proposition 8.1.

b) Note the analogy between the spectral sequence of Proposition 8.1 and Leray's spectral sequence for a covering (X_i) of Y, here by X_i étale over Y. The latter would be obtained formally by writing the spectral sequence of Proposition 8.1 for $X = \coprod_i X_i$. It is likely that these two spectral sequences admit a common generalization, valid whenever one has a family of morphisms $(X_i \to Y)$ which is a universal effective descent family for the fibred category of étale sheaves over a variable base scheme; compare No. 9 below. The analogous question also arises in ordinary topology, concerning a common generalization of the two kinds of Leray spectral sequences for coverings, assumed either open or locally finite closed. Since these lines were written, P. Deligne has developed a general theory of descent spectral sequences; see Exposé V bis.

Proposition 8.4. Let Y be a scheme, let π be a profinite group, let $(\pi_i)_i$ be the projective system of its finite discrete quotient groups, let $(X_i)_{i\in I}$ be a projective system of principal coverings of Y, with groups π_i, the transition homomorphisms $X_j \to X_i$ being compatible with the homomorphisms $\pi_j \to \pi_i$ on the operator groups, and put $X = \varprojlim X_i$ (cf. Exposé VII, 5.1), so that π acts on the Y-scheme X. Let F be an abelian sheaf on Y. Then there is a Hochschild-Serre spectral sequence, functorial in F,

$$H^*(Y,F) \Longleftarrow E_2^{pq} = H^p\big(\pi, \varinjlim_i H^q(X_i, F_{X_i})\big),$$

where F_{X_i} is the inverse image of F on X_i, and $H^p(\pi,-)$ denotes Galois cohomology.

The term E_2^{pq} displayed here is also, by definition of $H^p(\pi,-)$ and by transitivity of inductive limits, isomorphic to

$$E_2^{pq} = \varinjlim_i H^p(\pi_i, H^q(X_i, F_{X_i})).$$

It is therefore enough to find an inductive system of spectral sequences, depending on i,

$$H^*(Y,F) \Longleftarrow {}^iE_2^{pq} = H^p(\pi_i, H^q(X_i, F_{X_i})).$$

This reduces us to defining the spectral sequence when π is finite. Then $f : X \to Y$ is a covering morphism in $Y_{\text{ét}}$, so we may write the Leray spectral sequence of this morphism. Taking into account the canonical isomorphisms

$$X^n \simeq X \times^G G^n,$$

a well-known calculation shows that for every presheaf $\mathcal{H}$ on $X_{\text{ét}}$ which transforms sums into products, $C^*(X/Y, \mathcal{H})$ is precisely the complex of homogeneous cochains of G with coefficients in $\mathcal{H}(X)$. This gives the asserted form of the initial term.

A variant of this proof, avoiding any calculation, consists in viewing $E \mapsto X \times^\pi E$ as a morphism from the site of finite π-sets to the étale site of Y, and in writing the Leray spectral sequence of this morphism. Finally, when π is finite, the Hochschild-Serre spectral sequence can also be regarded as a special case of Proposition 8.1 relative to the morphism $f : X \to Y$.

Corollary 8.5. Assume Y quasi-compact and quasi-separated. Then the spectral sequence of Proposition 8.4 is written

$$H^*(Y,F) \Longleftarrow E_2^{pq} = H^p(\pi, H^q(X, F_X)).$$

Indeed, by Exposé VII, 5.8, there are canonical isomorphisms

$$\varinjlim_i H^q(X_i, F_{X_i}) \simeq H^q(X, F_X).$$

Corollary 8.6. Let Y be a henselian local scheme with closed point y, let F be an abelian sheaf on Y, and let F_0 be the induced sheaf on $Y_0 = \operatorname{Spec} k(y)$. Then the canonical homomorphisms

$$H^n(Y,F) \longrightarrow H^n(Y_0, F_0)$$

are isomorphisms.

Indeed, let $\bar{y}$ be a geometric point over y, corresponding to a separable closure $k(\bar{y})$ of $k(y)$. Since Y is henselian, the strict localization X of Y at $\bar{y}$ is the projective limit of the connected pointed Galois étale coverings X_i. Thus the hypotheses of Corollary 8.5 apply. Since $H^q(X, F_X) = 0$ for $q > 0$ by Corollary 4.7, one obtains isomorphisms

$$H^n(Y, F) \xleftarrow{\sim} H^n(\pi, F(X)),$$

where π is the Galois group of X over Y, isomorphic to that of $k(\bar{y})$ over $k = k(y)$. Similarly, or by Corollary 2.3,

$$H^n(Y_0, F_0) \xleftarrow{\sim} H^n(\pi, F_0(X_0)),$$

where $X_0 = X \times_Y Y_0 \simeq \operatorname{Spec} k(\bar{y})$. The restriction homomorphism $F(X) \to F_0(X_0)$ is an isomorphism by Corollary 4.8, and the conclusion follows.

9. Descent of étale sheaves

This section will not be used later in the seminar and may be omitted on a first reading.

Proposition 9.1. Let $f : X' \to X$ be a surjective, respectively universally submersive (SGA 1, IX, 2.1), morphism of schemes. Then the functor $f^* : X_{\text{ét}} \to X'_{\text{ét}}$ is faithful and conservative (cf. Corollary 3.6), respectively induces a fully faithful functor from the category of sheaves on $X_{\text{ét}}$ to the category of sheaves on $X'_{\text{ét}}$ endowed with descent data relative to $f : X' \to X$.

The first point follows immediately from the corresponding assertions on the fibres (3.6, 3.7). The second, which can also be stated by saying that f is a descent morphism relative to the fibred category of étale sheaves over variable schemes, means that for two sheaves F, G on X, the natural diagram

$$\operatorname{Hom}(F, G) \longrightarrow \operatorname{Hom}(F', G') \rightrightarrows \operatorname{Hom}(F'', G'') \qquad (*)$$

is exact, where F' and G', respectively F'' and G'', are the inverse images of F and G on X', respectively on $X'' = X' \times_X X'$. Taking F to be the final sheaf gives:

Corollary 9.2. Let $f : X' \to X$ be universally submersive. Then for every sheaf F on X, the diagram

$$\Gamma(X, F) \longrightarrow \Gamma(X', F') \rightrightarrows \Gamma(X'', F'')$$

is exact, where $X'' = X' \times_X X'$, and F', F'' are the inverse images of F on X', X''.

We prove Proposition 9.1 using the following lemma.

Lemma 9.3. Assume f universally submersive. Let F be a sheaf on X, denote by $S(F)$ the set of subsheaves of F, and define $S(F')$, $S(F'')$ similarly. Then the natural diagram of maps

$$S(F) \longrightarrow S(F') \rightrightarrows S(F'')$$

is exact.

Proof. The injectivity of $S(F) \to S(F')$ follows immediately from the fact that f^* is conservative. Indeed, if F_i, $i = 1, 2$, are two subsheaves of F such that $f^*(F_1) = f^*(F_2)$, then the inclusions $F_3 = F_1 \cap F_2 \to F_i$ are isomorphisms after applying f^*, hence are isomorphisms, and so $F_1 = F_2$.

It remains to prove that if G' is a subsheaf of F' such that $\mathrm{pr}_1^*(G') = \mathrm{pr}_2^*(G')$, then G' is the inverse image of a subsheaf of F. Choose an epimorphism $F_1 \to F$ in $X_{\text{ét}}$, with F_1 representable by an étale scheme over X. Let $F_2 = F_1 \times_F F_1$, and similarly define F_1', F_1'', F_2', F_2''. These sheaves give a diagram of sets

$$\begin{array}{ccccc}
S(F) & \longrightarrow & S(F') & \longrightarrow & S(F'') \\
\downarrow & & \downarrow & & \downarrow \\
S(F_1) & \longrightarrow & S(F_1') & \rightrightarrows & S(F_1'') \\
\downarrow\downarrow & & \downarrow\downarrow & & \downarrow\downarrow \\
S(F_2) & \longrightarrow & S(F_2') & \rightrightarrows & S(F_2''),
\end{array}$$

in which the columns are exact by the exactness properties in the categories of sheaves on X, X', and X'', respectively. A standard diagram chase shows that, to prove exactness of the first row, it is enough to prove it for the second and third rows. For the second row this follows from the fact that F_1 is representable, $X' \to X$ is universally submersive, and SGA 1, IX, 2.3. The sheaf F_2 is also representable: it is a subsheaf of $F_1 \times_X F_1$, which is representable, and one applies Proposition 6.1. Thus the third row is exact as well. $\square$

We now prove Proposition 9.1, namely that every morphism $u' : F' \to G'$, compatible with the descent data on F' and G', comes from a morphism $u : F \to G$. Put $H = F \times G$, so $H' = F' \times G'$ and $H'' = F'' \times G''$. The graph of u' is a subsheaf Γ' of H', whose two inverse images are equal to the graph of the same morphism $u'' : F'' \to G''$. Hence, by Lemma 9.3, Γ' comes from a subsheaf Γ of H. I claim that Γ is the graph of a morphism $u : F \to G$, i.e. that the morphism $p : \Gamma \to F$ induced by pr_1 is an isomorphism. Indeed, it becomes an isomorphism after base change by $X' \to X$, and we apply the part of Proposition 9.1 already proved. The morphism $u : F \to G$ then has the desired property. $\square$

Theorem 9.4. Let $f : X' \to X$ be a surjective morphism of schemes. Assume that one of the following hypotheses holds:

a) f is integral;

b) f is proper;

c) f is flat and locally of finite presentation;

d) X is discrete, for example the spectrum of a field.

Then f is an *effective descent morphism* for the fibred category $\mathcal{F}$ over (Sch) of étale sheaves on variable schemes. That is, f^* induces an equivalence from the category of sheaves on X to the category of sheaves on X' endowed with descent data relative to $f : X' \to X$.

9.4.1

Let F' be a sheaf on X', endowed with descent data relative to $f : X' \to X$. Define a functor

$$G : X^\circ_{\text{ét}} \longrightarrow (\mathbf{Set})$$

by

$$G(Y) = \mathrm{Ker}\big(F'(Y') \rightrightarrows F''(Y'')\big).$$

One immediately checks that G is a sheaf for the étale topology. Moreover there is an injective canonical homomorphism $G \to f_*(F')$, hence a homomorphism $f^*(G) \to F'$, and the latter is evidently compatible with descent data. It remains to examine whether this homomorphism is an isomorphism.

Note that G is also defined by exactness of

$$G \longrightarrow f_*(F') \rightrightarrows g_*(F''),$$

so for every base change $Y \xrightarrow{h} X$ which commutes with the formation of $f_*(F')$ and $g_*(F'')$, the sheaf $h^*(G) = G_Y$ identifies with the sheaf descended from $F'_{Y'}$ by $Y' \to Y$; and of course the homomorphism $f_Y^*(G_Y) \to (F_Y)'$ is obtained from $f^*(G) \to F'$ by base change. Now assume that the base change $Y = X' \to X$ commutes with the formation of $f_*(F')$ and $g_*(F'')$. Since $Y' = X' \times_X X'$ has a section over $Y = X'$, the morphism $Y' \to Y$ is an effective descent morphism for every fibred category, in particular for $\mathcal{F}$. It follows that $f_Y^*(G_Y) \to (F_Y)'$ is an isomorphism, and therefore $f^*(G) \to F'$ becomes an isomorphism after base change by $Y' = X' \times_X X' \to X'$. As the latter morphism has a section, $f^*(G) \to F'$ is itself an isomorphism. Thus the descent datum on F' is effective.

9.4.2

This proves Theorem 9.4 in case (a), by Corollary 5.6. Case (b) follows similarly from the fact that if f is proper then f_* commutes with every base change $Y' \to Y$; this will be proved in a later exposé, XII, 5.1(i).

9.4.3

In case (c), the question is local on X. We may assume X affine and that there exists an affine scheme X_1', of finite presentation, quasi-finite and faithfully flat over X, together with an X-morphism $X_1' \to X'$. This reduces us to the case where f is quasi-finite. After further localization on X, this time in the étale topology, there exists an open X_1' of X' such that $X_1' \to X$ is finite and surjective, reducing us to the finite surjective case already treated in (a).

9.4.4

In case (d), after localizing on X, we may suppose that X is a single point, and, taking Theorem 1.1 into account, even that it is the spectrum of a field k. Moreover f is universally open (EGA IV, 2.4.9), so the hypotheses of Proposition 9.1 apply, and the argument of 9.4.1 still applies, with a base change by $Y = \operatorname{Spec} k(x)$, for $x \in X$. In that case it is still true that f_* commutes with the base change $Y \to X$, as will be seen later (XVI, 1.4 and 1.5).

This completes the proof of Theorem 9.4.

SGA 4, Exposé IX

Constructible Sheaves; Cohomology of an Algebraic Curve

M. Artin

0. Introduction

Starting with the present exposé, in contrast with the preceding exposés and with the classical cohomological theory of topological spaces, we are forced, for the validity of the essential statements, to restrict to *torsion* sheaves, and often more precisely to torsion sheaves prime to the residual characteristics.

Let us note that in fact a "reasonable" theory of cohomology with integral, or even real, coefficients for varieties over an algebraically closed field of characteristic $p \neq 0$, analogous to the classical theory for $k = \mathbb{C}$, does not exist, as Serre observed. More precisely, there is no contravariant functor H^1, defined for instance on the category of smooth projective schemes over an algebraically closed field K of characteristic $p > 0$, with values in finite-dimensional vector spaces over $\mathbb{R}$, or over a subfield of $\mathbb{R}$, commuting with products, i.e. such that

$$H^1(X \times Y) \xleftarrow{\sim} H^1(X) \times H^1(Y),$$

and such that $\dim H^1(X) = 2$ when X is a connected curve of genus 1. Indeed, it would follow that, if X is an abelian variety over k, the map $u \mapsto u^1$ from $\operatorname{End}(X)$ to $\operatorname{End}(H^1(X))$ is additive, hence is a representation of the opposite ring of $\operatorname{End}(X)$. But in characteristic p, there exist elliptic curves X whose endomorphism ring is a maximal order A in a quaternion algebra over $\mathbb{Z}$ ([3], p. 198), and such an algebra plainly has no representation on a vector space of dimension 2 over $\mathbb{R}$, since $A \otimes_{\mathbb{Z}} \mathbb{R}$ is a division ring, namely Hamilton's quaternions.

1. The formalism of torsion sheaves

Let T be a topos and F an abelian sheaf on T. We can multiply by n in F, for $n \in \mathbb{Z}$; this multiplication is induced by the analogous operation in $(\mathbf{Ab})$. Denote by ${}_nF$ the kernel of this multiplication. Thus ${}_nF(X)$, for $X \in \operatorname{Ob} T$, is the group of sections of $F(X)$ whose order divides n. Let $\mathcal{P}$ denote the set of all prime numbers.

Definition 1.1. Let p be a set of prime numbers. We say that F is a sheaf of p-torsion, or a p-sheaf, if the canonical morphism

$$\varinjlim_n {}_nF \longrightarrow F$$

is bijective, where n runs through the positive integers such that $\operatorname{ass}(n) \subset p$, i.e. such that all prime divisors of n belong to p. If $p = \mathcal{P}$ is the set of all prime numbers, we simply say that F is torsion.

Proposition 1.2.

(i) The sheaf F is of p-torsion if and only if F is the sheaf associated with a presheaf with values in abelian groups of p-torsion. One may take

$$P = \overset{\text{presheaves}}{\varinjlim_{\operatorname{ass}(n) \subset p}} {}_nF.$$

(ii) If F is of p-torsion and $X \in \operatorname{Ob} T$ is quasi-compact, then $F(X)$ is a group of p-torsion.

(iii) If T is locally of finite type (VI, 1.1), then F is of p-torsion if and only if $F(X)$ is a group of p-torsion for each quasi-compact $X \in \operatorname{Ob} T$. In this case one has

$$\varinjlim_{\operatorname{ass}(n) \subset p} H^q(T/X; {}_nF) \xrightarrow{\sim} H^q(T/X; F)$$

for every quasi-compact X and every q.

(iv) If $u : T \to T'$ is a morphism of topoi and if F' is of p-torsion on T', then the inverse image u^*F' is a sheaf of p-torsion on T.

(v) If $u : T \to T'$ is a morphism of topoi, with T and T' locally of finite type and u quasi-compact, and if F is a sheaf of p-torsion on T, then the sheaves $R^q u_* F$ are of p-torsion on T', for all q.

Proof. For (i), if a sheaf F is of p-torsion, then F is plainly the sheaf associated with the presheaf P whose group of sections $P(X)$ is the subgroup of p-torsion elements of $F(X)$. Conversely, let $F = aP$ be the sheaf associated with P, where P takes values in abelian groups of p-torsion. From the exact sequence

$$0 \longrightarrow {}_nP \longrightarrow P \xrightarrow{n} P$$

one obtains an exact sequence

$$0 \longrightarrow a({}_nP) \longrightarrow F \xrightarrow{n} F,$$

hence $a({}_nP) = {}_nF$. Since the functor a commutes with inductive limits, the isomorphism

$$\varinjlim_{\mathrm{ass}(n)\subset p}^{\text{presheaves}} {}_nP \xrightarrow{\sim} P$$

gives

$$\varinjlim_{\mathrm{ass}(n)\subset p} {}_nF \xrightarrow{\sim} F.$$

For (ii), put $P = \varinjlim^{\text{presheaves}} {}_nF$. Then $P \subset F$, so P is a separated presheaf, and

$$aP(X) = \varinjlim \ker\Big(\prod_i P(X_i) \rightrightarrows \cdots\Big),$$

where, as usual, the limit is taken over the covering families $\{X_i \to X\}$. Since X is quasi-compact, it is enough to use finite covering families. For such families, $\prod_i P(X_i)$ is of p-torsion, and hence so is its kernel. Thus $aP(X) = F(X)$ is of p-torsion.

For (iii), to check that $\varinjlim {}_nF \to F$ is bijective, it is enough to check values on each X in a family of generators. Thus the first assertion reduces to (ii). The displayed isomorphism follows from the same passage to inductive limits in cohomology for locally finite type topoi. Assertion (iv) follows from (i), since $u^*(aP') = a(u^\bullet P')$. Assertion (v) is an easy consequence of (iii).

We shall also need a non-abelian variant. There are technical difficulties in the definition, and we content ourselves with giving the definition for the étale topology of a scheme.

Definition 1.3. Let p be a set of prime numbers. A group G is called an *ind-p-group* if every finite subset of G generates a finite subgroup of order n, with $\mathrm{ass}(n) \subset p$. Equivalently, the finite subgroups G_α of order n_α, with $\mathrm{ass}(n_\alpha) \subset p$, form a filtered system and $G = \varinjlim G_\alpha$. If $p = \mathcal{P}$, one says *ind-finite group* instead of ind-p-group.

One verifies immediately:

Proposition 1.4.

(i) A subgroup and a quotient of an ind-p-group is again an ind-p-group.

(ii) A filtered inductive limit of ind-p-groups is an ind-p-group.

(iii) A finite projective limit of ind-p-groups is an ind-p-group.

Definition 1.5. Let X be a scheme. A sheaf F of groups on X is called a *sheaf of ind-p-groups*, respectively a sheaf of *ind-finite groups*, if for every étale $U \to X$, with U quasi-compact, the group $F(U)$ is an ind-p-group, respectively an ind-finite group.

Proposition 1.6. Let F be a sheaf of groups on a scheme X.

(i) The sheaf F is a sheaf of ind-p-groups if and only if for every geometric point ξ of X, the fibre F_ξ is an ind-p-group.

(ii) If $f : X \to X'$ is a morphism and F' is a sheaf of ind-p-groups on X', then f^*F' is a sheaf of ind-p-groups.

(iii) If $f : X \to X'$ is quasi-compact and F is a sheaf of ind-p-groups on X, then f_*F is a sheaf of ind-p-groups.

Proof. For (i), suppose first that F is a sheaf of ind-p-groups, and let ξ be a geometric point of X. Then $F_\xi = \varinjlim F(X')$, where X' runs through a pseudo-filtered system of schemes étale over X (VIII, 3.9). The affine, hence quasi-compact, X' form a cofinal system, so F_ξ is an inductive limit of ind-p-groups, hence an ind-p-group by Proposition 1.4(ii). Conversely, assume that every fibre F_ξ is an ind-p-group, and let $U \to X$ be étale with U quasi-compact. Let $S \subset F(U)$ be finite. For every geometric point ξ of U, the image of S in F_ξ generates a finite group. Since a finite group is finitely presented, one easily concludes that there exists an étale U_ξ over U, pointed by ξ, such that S generates a finite group in $F(U_\xi)$ (cf. VIII, 4). A finite number of such U_ξ cover U, say $\{U_1, \dots, U_n\}$. Then the image of S in $\prod F(U_i)$ generates a finite subgroup, and since $F(U) \subset \prod F(U_i)$, the conclusion follows. Assertion (ii) is trivial from (i), and (iii) is immediate.

3. Kummer and Artin–Schreier theories

3.0

We denote by $(\mathbb{G}_m)_X$ the sheaf represented on X by the multiplicative group $\operatorname{Spec} \mathcal{O}_X[t, t^{-1}]$. Thus, for an *étale* morphism $U \to X$, its sections over U are the units $\Gamma(U, \mathcal{O}_U)^*$. Let $n \in \mathbb{N}$. The kernel of the n-th power map on $(\mathbb{G}_m)_X$ is the sheaf of n-th roots of unity, denoted $(\mu_n)_X$. There is therefore an exact sequence at the left

$$0 \longrightarrow (\mu_n)_X \longrightarrow (\mathbb{G}_m)_X \xrightarrow{n} (\mathbb{G}_m)_X. \tag{3.1}$$

If n is invertible on X, that is, prime to all residual characteristics of X, the n-th power map is surjective as a morphism of *étale* sheaves:

Kummer theory 3.2. If n is invertible on X, then

$$0 \longrightarrow (\mu_n)_X \longrightarrow (\mathbb{G}_m)_X \xrightarrow{n} (\mathbb{G}_m)_X \longrightarrow 0$$

is exact.

Indeed, for $u \in (\mathbb{G}_m)_X(U)$, with $U \to X$ *étale*, one must find an *étale* covering on which u becomes an n-th power. Since n is invertible on U, the equation

$$T^n - u = 0$$

is separable over U, so

$$U' = \operatorname{Spec} \mathcal{O}_U[T]/(T^n - u)$$

is *étale* over U. The morphism $U' \to U$ is surjective, hence covering, and on U' the section u is an n-th power.

When n is invertible on X, the sheaf $(\mu_n)_X$ is locally constant, in fact locally isomorphic to $(\mathbb{Z}/n\mathbb{Z})_X$, and is constant in important cases. Kummer theory therefore gives information about the cohomology of X with values in certain constant sheaves. By definition

$$H^0(X, (\mathbb{G}_m)_X) = \Gamma(X, \mathcal{O}_X^*),$$

and in degree one one has the following form of Hilbert's Theorem 90.

Theorem 3.3. There is an isomorphism

$$H^1(X, (\mathbb{G}_m)_X) \xrightarrow{\sim} \operatorname{Pic} X,$$

where $\operatorname{Pic} X$ is the group of isomorphism classes of invertible sheaves on X.

Equivalently, the canonical morphism

$$H^1(X_{\text{Zar}}, (\mathbb{G}_m)_X) \longrightarrow H^1(X_{\text{ét}}, (\mathbb{G}_m)_X)$$

is bijective. This amounts to saying that

$$R^1\epsilon_*(\mathbb{G}_m)_X = 0,$$

where $\epsilon : X_{\text{ét}} \to X_{\text{Zar}}$ is the evident morphism of sites (VIII, 4). Thus one is reduced to the affine case. Since H^1 may be computed by Čech cohomology (V, 2.5), and since for quasi-compact X it is enough to use quasi-compact coverings, the assertion follows at once from descent theory for invertible sheaves (cf. FGA, p. 190, or SGA 1, VIII, 1).

3.3.1 Cartier divisors

Assume that X is noetherian and has no embedded components, and let x_j be the maximal points of X. Let R_j be the local artinian ring of X at x_j, and let $i_j : \operatorname{Spec} R_j \to X$ be the inclusion. There is a canonical injection, both on $X_{\text{ét}}$ and on X_{Zar},

$$(\mathbb{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbb{G}_m)_{\operatorname{Spec} R_j}.$$

Its cokernel D is called the sheaf of Cartier divisors on $X_{\text{ét}}$, respectively on X_{Zar}. In EGA IV, 21, these are simply called divisors.

Corollary 3.4. Let X be noetherian and without embedded components, let $\epsilon : X_{\text{ét}} \to X_{\text{Zar}}$ be the canonical morphism of sites, and let D be the sheaf of Cartier divisors on $X_{\text{ét}}$. Then $\epsilon_* D$ is the sheaf D_{Zar} of Cartier divisors on X_{Zar}. In particular

$$H^0(X_{\text{ét}}, D) \simeq H^0(X_{\text{Zar}}, D_{\text{Zar}}).$$

This follows by applying ϵ_* to the exact sequence

$$0 \longrightarrow (\mathbb{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbb{G}_m)_{\operatorname{Spec} R_j} \longrightarrow D \longrightarrow 0,$$

which remains exact because $R^1\epsilon_*(\mathbb{G}_m)_X = 0$ by Theorem 3.3.

In characteristic $p \neq 0$, the following result will sometimes replace Kummer theory for the study of p-torsion coefficients.

Artin–Schreier theory 3.5. If X is of characteristic $p > 0$, there is an exact sequence

$$0 \longrightarrow (\mathbb{Z}/p\mathbb{Z})_X \longrightarrow \mathcal{O}_X \xrightarrow{\wp} \mathcal{O}_X \longrightarrow 0,$$

where $\wp$ is the morphism of additive groups defined by $\wp(f) = f^p - f$.

The kernel of $\wp$ is the sheaf of sections of $\mathcal{O}_X$ which locally lie in the prime field, hence is constant. The map $\wp$ is surjective for the *étale* topology, since the equation $T^p - T - a$, with $a \in \Gamma(U, \mathcal{O}_U)$, is always separable in characteristic p, hence defines an *étale* covering of U. In applying this exact sequence one uses VII, 4.2.

Proposition 3.6.

(i) Let X be a scheme and let $i : x \to X$ be the inclusion of a point. Then

$$H^1(X, i_*(\mathbb{Z})_x) = 0,$$

where $(\mathbb{Z})_x$ denotes the constant sheaf of value $\mathbb{Z}$ on the point x.

(ii) If X is irreducible and if, for every geometric point $\bar{x}$ of X, the strict localization $X_{\bar{x}}$ is irreducible, in other words if X is geometrically unibranch, then

$$H^1(X, (\mathbb{Z})_X) = 0.$$

Proof. For (i), the Leray spectral sequence

$$E_2^{pq} = H^p(X, R^q i_*(\mathbb{Z})_x) \Longrightarrow H^{p+q}(x, (\mathbb{Z})_x)$$

gives an injection $H^1(X, i_*(\mathbb{Z})_x) \to H^1(x, (\mathbb{Z})_x)$. But, since $x = \operatorname{Spec} k(x)$, the group $H^1(x, (\mathbb{Z})_x)$ is zero: this follows from the fact that $H^q(x, F)$, for $q > 0$, is torsion (VII, 2.2 and CG), and from the exact sequences

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

For (ii), let $i : x \to X$ be the inclusion of the generic point. Under the stated hypotheses the canonical morphism $(\mathbb{Z})_X \to i_*(\mathbb{Z})_x$ is immediately seen to be bijective, and the assertion follows from (i). The converse to this bijectivity is also true.

Remarks 3.7. In Proposition 3.6(i) and (ii), $\mathbb{Z}$ may be replaced by any torsion-free abelian group. On the other hand, (ii) becomes false if the geometrically unibranch hypothesis is dropped; for example take an algebraic curve over a field with an ordinary double point and compute $H^1(X, \mathbb{Z})$.

4. The case of an algebraic curve

4.0

Let X be a noetherian scheme of dimension one. Let $R(X)$ be the ring of rational functions on X, and let $i : \operatorname{Spec} R(X) \to X$ be the inclusion. The ring $R(X)$ is artinian: it is the product of the local rings of X at its maximal points, so one may use the results of VIII, 2.

Let F be an abelian sheaf on $\operatorname{Spec} R(X)$, and consider the sheaves $R^q i_* F$, $q > 0$, on X. It follows at once from VIII, 5.2 that the fibre of $R^q i_* F$ at a geometric point lying over a maximal point of X is zero. Since X has dimension one, such a sheaf is of a very special kind.

Lemma 4.1. Let X be noetherian and let F be a sheaf on X. The following conditions are equivalent; a sheaf satisfying them will be called a skyscraper sheaf.

(i) The fibre of F at every geometric point $\bar{x}$ lying over a non-closed point $x \in X$ is zero.

(ii) Each section $s \in F(X')$, with $X' \to X$ *étale* of finite type, is zero except at finitely many closed points of X'.

(iii) One has an isomorphism

$$F \simeq \bigoplus_x i_{x*} i_x^* F,$$

where x runs through the closed points of X, and $i_x : x \to X$ is the inclusion.

Proof. If (i) holds and $z \in F(X')$, its support is a closed subset of X' formed of closed points, hence is finite because X' is noetherian; this proves (ii). Conversely, (ii) implies (i) by the calculation of fibres (VIII, 3.9). Condition (iii) plainly implies (i), since the functor of fibres commutes with direct sums and the support of $i_{x*} G_x$ is contained in $\{x\}$. Finally, assuming (ii), the canonical morphism

$$F \longrightarrow \prod_x i_{x*} i_x^* F$$

factors through the direct sum; the resulting morphism to $\bigoplus_x i_{x*} i_x^* F$ is checked directly to be bijective.

4.1.1

Applying the lemma to the sheaf $R^q i_* F$ introduced above gives

$$R^q i_* F \simeq \bigoplus_x i_{x*} i_x^* (R^q i_* F),$$

with x running through the closed points of X. By VIII, 5, the fibre of $R^q i_* F$ at a geometric point $\bar{x}$ of X is

$$(R^q i_* F)_{\bar{x}} \simeq H^q(\operatorname{Spec} R(X_{\bar{x}}), F),$$

where $X_{\bar{x}}$ is the strict localization of X at $\bar{x}$, and F also denotes the induced sheaf on $\operatorname{Spec} R(X_{\bar{x}})$. Assume now that the residue field of every closed point of X is separably closed. Then

$$H^p(X, R^q i_* F) \simeq \begin{cases} \bigoplus_x H^q(\operatorname{Spec} R(X_x), F), & p = 0, \\ 0, & p > 0, \end{cases}$$

for $q > 0$, where X_x denotes the strict localization at x.

Corollary 4.2. Let X be a noetherian scheme of dimension one such that, for every closed point x, the residue field $k(x)$ is separably closed. Let F be an abelian sheaf on $\operatorname{Spec} R(X)$. In the Leray spectral sequence

$$E_2^{pq} = H^p(X, R^q i_* F) \Longrightarrow H^{p+q}(\operatorname{Spec} R(X), F)$$

one has

$$E_2^{pq} = 0 \quad (p > 0, q > 0), \qquad E_2^{0q} = \bigoplus_x H^q(\operatorname{Spec} R(X_x), F) \quad (q > 0).$$

Consequently there is an exact sequence

$$\begin{aligned} 0 \to H^1(X, i_*F) \to H^1(\operatorname{Spec} R(X), F) \to \bigoplus_x H^1(\operatorname{Spec} R(X_x), F) \to H^2(X, i_*F) \to \\ \to H^2(\operatorname{Spec} R(X), F) \to \bigoplus_x H^2(\operatorname{Spec} R(X_x), F) \to \cdots . \end{aligned}$$

4.3

Assume further that the ℓ-cohomological dimension of $\operatorname{Spec} R(X)$ is at most one. Let $K^1, \dots, K^m$ be the residue fields of the artinian ring $R(X)$, equivalently the residue fields at the maximal points of X. By the dictionary of VIII, 2,

$$\operatorname{cd}_\ell(\operatorname{Spec} R(X)) = \sup_i \operatorname{cd}_\ell(G(\overline{K}^i/K^i)).$$

Every residue field K_x of $R(X_x)$ is a separable extension, possibly infinite, of one of the K^i, hence

$$\operatorname{cd}_\ell(G(\overline{K}_x/K_x)) \le \operatorname{cd}_\ell(G(\overline{K}^i/K^i))$$

by CG II, 4.1, Proposition 10. Applying Corollary 4.2 to an ℓ-torsion sheaf F, one obtains

$$\begin{cases} 0 \to H^1(X, i_*F) \to H^1(\operatorname{Spec} R(X), F) \to \bigoplus_x H^1(\operatorname{Spec} R(X_x), F) \to H^2(X, i_*F) \to 0, \\ H^q(X, i_*F) = 0 \quad (q > 2). \end{cases} \tag{4.3.1}$$

The hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \le 1$ is satisfied, for example, when X is an algebraic curve over a separably closed field, by Tsen's theorem. Exact sequences of this type were studied by Ogg and Shafarevich.

4.4

Assume that ℓ is invertible on X, and take $F = (\mathbb{G}_m)_{R(X)}$. From Theorem 3.3 and CG, Chapter I, Proposition 12, one has, under the same hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \le 1$,

$$H^1(\operatorname{Spec} R(X), \mathbb{G}_m) = 0, \qquad H^q(\operatorname{Spec} R(X), \mathbb{G}_m) \text{ has no } \ell\text{-torsion for } q \ge 2. \tag{4.4.1}$$

The same holds with $R(X)$ replaced by $R(X_x)$. Corollary 4.2 therefore gives

$$H^1(X, i_*(\mathbb{G}_m)_{R(X)}) = 0, \qquad H^q(X, i_*(\mathbb{G}_m)_{R(X)}) \text{ has no } \ell\text{-torsion for } q > 1. \tag{4.4.2}$$

For any sheaf F on X, Lemma 4.1 shows that the kernel and cokernel of the canonical homomorphism $F \to i_*i^*F$ are skyscraper sheaves. Since the closed residue fields are separably closed, these skyscraper sheaves have no higher cohomology. Thus

$$H^i(X, F) \longrightarrow H^i(X, i_*i^*F)$$

is an isomorphism for $i \geq 2$. Applied to $(\mathbb{G}_m)_X$, this yields

$$H^i(X, \mathbb{G}_m) \text{ has no } \ell\text{-torsion for } i \geq 2, \tag{4.5}$$

under the hypotheses of (4.4). Combining this with Kummer theory and Theorem 3.3 gives:

Theorem 4.6. Let X be noetherian of dimension one, let n be an integer invertible on X, suppose that $\mathrm{cd}_\ell(R(X)) \leq 1$ for every prime ℓ dividing n, and suppose that for every closed point x of X the residue field $k(x)$ is separably closed. Then

$$H^q(X, (\mu_n)_X) = 0 \quad (q > 2),$$

and for $q \leq 2$ the cohomology is given by the exact sequence

$$\begin{aligned} 0 \to H^0(X, (\mu_n)_X) \to \Gamma(X, \mathcal{O}_X^*) \xrightarrow{n} \Gamma(X, \mathcal{O}_X^*) \to H^1(X, (\mu_n)_X) \\ \to \operatorname{Pic} X \xrightarrow{n} \operatorname{Pic} X \to H^2(X, (\mu_n)_X) \to 0. \end{aligned}$$

Corollary 4.7. Let X be a complete connected curve over a separably closed field k, of characteristic prime to n. Then

$$H^0(X, (\mu_n)_X) \simeq \mu_n(k),$$
$$H^1(X, (\mu_n)_X) \simeq {}_n \operatorname{Pic} X,$$

and

$$H^2(X, (\mu_n)_X) \simeq (\operatorname{Pic} X)/n \simeq (\mathbb{Z}/n\mathbb{Z})^c,$$

where c is the number of irreducible components of X.

For the last assertion, recall that the Picard scheme of such an X/k has a decomposition

$$0 \longrightarrow A \longrightarrow \mathcal{P}ic_{X/k} \longrightarrow \mathbb{Z}^c \longrightarrow 0, \tag{4.8}$$

where A is a connected algebraic group over k. Multiplication by n on such an A is *étale*, hence surjective on k-valued points. Thus ${}_n(\operatorname{Pic} X) \simeq {}_nA(k)$ and $(\operatorname{Pic} X)/n \simeq (\mathbb{Z}/n\mathbb{Z})^c$. If X is smooth over k, then A is the Jacobian and ${}_n \operatorname{Pic} X \simeq (\mathbb{Z}/n\mathbb{Z})^{2g}$, where g is the genus. Since $\mu_n(k) \simeq \mathbb{Z}/n\mathbb{Z}$, the ranks of the cohomology groups are $1, 2g, 1$, as expected.

5. The trace method

5.1

Let X be a scheme, $f : X' \to X$ a finite *étale* covering, and F a sheaf on X. Adjunction gives a restriction morphism

$$F \xrightarrow{\text{res}} f_* f^* F.$$

Since f is finite, $H^q(X, f_* f^* F) \simeq H^q(X', f^* F)$, whence the usual restriction map

$$H^q(X, F) \xrightarrow{\text{res}} H^q(X', f^* F). \tag{5.1.1}$$

There is also a trace morphism

$$f_* f^* F \xrightarrow{\text{tr}} F, \tag{5.1.2}$$

which induces

$$H^q(X', f^*F) \xrightarrow{\mathrm{tr}} H^q(X, F). \tag{5.1.3}$$

It is characterized by being local for the *étale* topology and, when X' is the disjoint sum of d copies of X, by being the summation map $F^d \to F$. The uniqueness is immediate, since every finite *étale* covering is locally constant; existence follows by the usual descent argument. Hence

$$\mathrm{tr} \circ \mathrm{res} : F \to F$$

is multiplication by the local degree of f. If the degree is a constant d, the same is true on cohomology.

Corollary 5.2. Let $f : X' \to X$ be a finite *étale* covering of constant degree d. If F is a sheaf of r-torsion on X, with $(r, d) = 1$, then $H^q(X', f^*F) = 0$ implies $H^q(X, F) = 0$.

Indeed, multiplication by d is an isomorphism of F, hence of its cohomology, and the resulting isomorphism factors through zero.

5.2.1

Let $i : U \to X$ be an immersion, let $f : U' \to U$ be a finite *étale* covering of degree d, and suppose that f is induced by a finite morphism $g : X' \to X$ by base change:

$$\begin{tikzcd} U' \arrow[r, "i'"] \arrow[d, "f"'] & X' \arrow[d, "g"] \\ U \arrow[r, "i"] & X. \end{tikzcd}$$

For a sheaf F on U one has the composites

$$F \xrightarrow{\mathrm{res}} f_* f^* F \xrightarrow{\mathrm{tr}} F,$$

whose composite is multiplication by d. Since $i_* f_* = g_* i'_*$, this gives

$$i_* F \xrightarrow{\mathrm{res}} g_* i'_* f^* F \xrightarrow{\mathrm{tr}} i_* F, \tag{5.3}$$

and since $i_! f_* \simeq g_* i'_!$,

$$i_! F \xrightarrow{\mathrm{res}} g_* i'_! f^* F \xrightarrow{\mathrm{tr}} i_! F, \tag{5.4}$$

again with composite multiplication by d. Passing to cohomology gives the corresponding restriction–trace composites for $i_* F$ and $i_! F$.

Proposition 5.5. Let X be quasi-compact and quasi-separated, let ℓ be a prime, and let H be a semi-exact functor to **Ab**, defined on the category of ℓ-sheaves on X, and compatible with filtered inductive limits. The following are equivalent:

(i) $H = 0$.

(ii) $H(f_* i_!(\mathbb{Z}/\ell\mathbb{Z})) = 0$ for every finite morphism $f : Y \to X$, and every open immersion $i : U \to Y$ with U of finite presentation over X. If X is noetherian, one may restrict to integral Y.

Proof. By Corollary 2.7.2 and Proposition 2.5 it is enough to test H on sheaves $i_!G$, where $i : U \to X$ is the inclusion of a subscheme of finite presentation over X, and G is an isotrivial locally constant ℓ-sheaf on U. Choose a principal covering $U'' \to U$, with group $\mathfrak{g}$, on which G becomes constant. Let A be the ℓ-group of its values, and let $\mathfrak{h}$ be an ℓ-Sylow subgroup of $\mathfrak{g}$. If $U' = U''/\mathfrak{h}$, then the degree d of $U' \to U$ is prime to ℓ. Extending $U' \to U$ to a finite morphism $X' \to X$ (EGA IV, 8.12.6), and applying (5.4), one reduces to proving the vanishing for the pullback of G to U'. As an $\mathfrak{h}$-module, A admits a filtration whose successive quotients are $\mathbb{Z}/\ell\mathbb{Z}$ with trivial action; the assertion follows by semi-exactness.

Proposition 5.6. Let X be noetherian, let ℓ be a prime, and let φ be a function with values in $\mathbb{N}$, defined on the integral schemes finite over X. Suppose $\varphi(Y_1) \leq \varphi(Y_2)$ whenever there is an X-morphism $Y_1 \to Y_2$, and that the inequality is strict if Y_1 is a proper closed subscheme of Y_2. For finite Y over X, put $\varphi(Y) = \sup \varphi(Y_i)$, where the Y_i are the irreducible components of Y, with their reduced induced structures. The following are equivalent:

(i) For every finite Y over X and every ℓ-sheaf F on Y, one has $H^q(Y, F) = 0$ for $q > \varphi(\operatorname{Supp} F)$.

(ii) For every integral finite Y over X, one has $H^q(Y, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for $q > \varphi(Y)$.

Proof. Assuming (ii), apply Proposition 5.5 to each H^q, using the facts that cohomology commutes with inductive limits and with direct image under finite morphisms. One is reduced to the case where Y is integral finite over X, and $F = i_!(\mathbb{Z}/\ell\mathbb{Z})_U$ for a non-empty open immersion $i : U \to Y$. The exact sequence

$$0 \longrightarrow i_!(\mathbb{Z}/\ell\mathbb{Z})_U \longrightarrow (\mathbb{Z}/\ell\mathbb{Z})_Y \longrightarrow (\mathbb{Z}/\ell\mathbb{Z})_Z \longrightarrow 0,$$

with $Z = Y - U \neq Y$, and induction on φ, give the vanishing. The converse is immediate.

Corollary 5.7. Let X be an algebraic curve over a separably closed field, of characteristic different from ℓ. Then the cohomology of X with values in any ℓ-torsion sheaf F vanishes in degrees > 2. If X is affine, it vanishes in degrees > 1.

For finite integral $Y \to X$, put $\varphi(Y) = 2 \dim Y$, or $\varphi(Y) = \dim Y$ in the affine case. Proposition 5.6 reduces the assertion to the case $F = \mathbb{Z}/\ell\mathbb{Z}$ on an integral curve. Since $(\mathbb{Z}/\ell\mathbb{Z})_Y \simeq (\mu_\ell)_Y$, Theorem 4.6 gives the vanishing in degrees > 2. In the affine case one may pass to the normalization and then to a smooth compactification $\bar{X}$; at least one point is missing from $\bar{X}$, so $A(k) \to \operatorname{Pic} X$ is surjective, where A is the Jacobian of $\bar{X}$. As $A(k)$ is divisible by $\ell \neq \operatorname{char} k$, so is $\operatorname{Pic} X$; hence $H^2(X, \mu_\ell) = 0$ by Theorem 4.6.

Remark 5.8. For later reference, the proof of Proposition 5.5 actually proves the following. Let X be quasi-compact and quasi-separated, ℓ a prime, and C a strictly full subcategory of the category of constructible abelian ℓ-torsion sheaves, stable under direct summands and extensions. Suppose that for every finite morphism $f : Y \to X$ and every open immersion $i : U \to Y$ with U of finite presentation over X, one has

$$f_*(i_!(\mathbb{Z}/\ell\mathbb{Z})_U) \in C.$$

Then C contains all constructible ℓ-torsion sheaves.

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SGA 4, Exposé X

Cohomological Dimension: First Results

M. Artin

1. Introduction

Throughout this exposé, unless the contrary is expressly stated, the word "sheaf" means "abelian sheaf."

Let X be a scheme and let ℓ be a prime number. Recall that the *ℓ-cohomological dimension* of X, denoted $\operatorname{cd}_\ell X$, is the largest integer n, or ∞ if no such integer exists, such that $H^n(X, F) \neq 0$ for at least one ℓ-torsion sheaf F on X. If $X = \operatorname{Spec} A$, we write $\operatorname{cd}_\ell A = \operatorname{cd}_\ell \operatorname{Spec} A$. Taking the dictionary of VIII, 2 into account, this recovers the classical notion of Galois cohomological dimension in the case where $A = k$ is a field. Since plainly

$$\operatorname{cd}_\ell\Big(\coprod_i X_i\Big) = \sup_i \operatorname{cd}_\ell X_i,$$

the classical theory extends immediately to arbitrary artinian rings. This theory is set out in Serre's *Cohomologie galoisienne*, cited below as CG; we shall add a few refinements in Section 2.

For instance, let X be an irreducible algebraic variety of dimension n over a separably closed field k, and fix $\ell \neq \operatorname{char} k$. The theorem of Section 4 gives the bound $2n$, the "true" dimension suggested by the topological analogy when $k = \mathbb{C}$. This result is quite elementary from the theory for fields. In fact the bound $2n$ can be improved except in the complete case: there is a bound $2n-1$ whenever at least one point is missing from X. This is the stable bound in the sense that every X of dimension n contains an open subset of cohomological dimension $2n-1$. For affine varieties one obtains the still sharper bound n, a generalization of a Lefschetz theorem for hyperplane sections. These two latter assertions are deeper and will be proved only later, in XIV.

Cohomological dimension is not a local notion, unlike what happens for paracompact spaces. A henselian local ring with separably closed residue field has cohomological dimension zero, but after removing its closed point one may obtain arbitrarily large cohomological dimension, namely $2n-1$ in the most important cases when $n = \dim A$. This again agrees with the topological analogy supplied by the cohomology of $U - \{x\}$, where U is a small open neighbourhood of a point of a complex analytic space.

2. Auxiliary results on fields

Theorem 2.1. Let k'/k be a finitely generated extension of a field k, and let ℓ be a prime. Then

$$\operatorname{cd}_\ell k' \leq \operatorname{cd}_\ell k + \operatorname{trdeg}(k'/k).$$

If the inequality is strict and $\ell \neq \operatorname{char} k$, then $\ell = 2$, the field k is orderable, and k' is not orderable, i.e. -1 is a sum of squares in k'.

When $\ell = \operatorname{char} k$, one has $\operatorname{cd}_\ell k' \leq 1$ (CG II, 2.2), so the question of equality in Theorem 2.1 scarcely arises. Also $\operatorname{cd}_2 k = \infty$ if k is orderable: an algebraic closure $\bar{k}$ contains a subextension

$\widetilde{k}$, a maximal ordered subfield, with $[\bar{k} : \widetilde{k}] = 2$; plainly $\operatorname{cd}_2 \widetilde{k} = \infty$, and $\operatorname{cd}_2 \widetilde{k} \leq \operatorname{cd}_2 k$ by CG I, Proposition 14.

Proof. The inequality, and equality when $\operatorname{cd}_\ell k < \infty$, are due to Tate (CG II, 4.1.2). It remains to show that $\operatorname{cd}_\ell k' < \infty$ and $\operatorname{cd}_\ell k = \infty$ imply that k is orderable and $\ell = 2$. It is enough to treat (a) finite extensions and (b) $k' = k(x)$ with x transcendental over k. In case (a), after assuming the extension separable, the absolute Galois group of k' is an open subgroup of that of k. A theorem of Serre gives equality of ℓ-dimensions except when the larger group contains an element of order ℓ. An algebraically closed field has no finite-index subfield except in the case $\ell = 2$ and a real closed situation (Artin–Schreier), giving the claim. In case (b), take the henselization R of $\operatorname{Spec} k[x]$ at $x = 0$, with fraction field K. If $\operatorname{cd}_\ell k(x) < \infty$, then $\operatorname{cd}_\ell K < \infty$; the result follows from the next theorem applied to R.

Theorem 2.2. Let R be a henselian discrete valuation ring, with fraction field K and residue field k. Let ℓ be a prime. Then

$$\operatorname{cd}_\ell K = \operatorname{cd}_\ell k + 1$$

in the following two cases:

(i) $\ell \neq \operatorname{char} k$;

(ii) $\ell \neq \operatorname{char} K$, the field k is perfect, and $\operatorname{cd}_\ell k < \infty$.

This theorem was recently proved by Ax. If $\ell = \operatorname{char} K$, one always has $\operatorname{cd}_\ell K \leq 1$ (CG II, 2.2). What remains in unequal characteristic, when $\operatorname{char} k = \ell$ and k is imperfect, should be related to the absolute module of differentials Ω^1_k (EGA IV, 20.4.3).

Lemma 2.2.1. Let R be a henselian discrete valuation ring with fraction field K, and let $\widehat{K}$ be the fraction field of the completion $\widehat{R}$. Then

$$G(\overline{\widehat{K}}/\widehat{K}) \simeq G(\bar{K}/K).$$

Proof. Equivalently, the functor $L \mapsto L \otimes_K \widehat{K}$ from finite *étale* K-algebras to finite *étale* $\widehat{K}$-algebras is an equivalence. Essential surjectivity is obtained by approximating a separable irreducible equation over $\widehat{R}$ by one over R, using the usual Newton method. For full faithfulness it is enough to see that if K' is a field, then $K' \otimes_K \widehat{K}$ is a field. If R' is the normalization of R in K', then R' is finite over R, a discrete valuation ring, and its completion is $R' \otimes_R \widehat{R}$; the corresponding total ring of fractions is therefore again a field.

The proof of Theorem 2.2 is then the standard argument of Tate, together with the preceding lemma. In case (i), when $\operatorname{cd}_\ell k < \infty$, one reduces to the complete case in CG II, Proposition 12. To see that $\operatorname{cd}_\ell K < \infty$ implies $\operatorname{cd}_\ell k < \infty$, let $\widetilde{K}$ be the maximal unramified extension of K, with residue field separably closed, and put $G = G(\bar{K}/K)$, $H = G(\bar{K}/\widetilde{K})$, $\bar{G} = G/H = G(\bar{k}/k)$. Passing to an ℓ-Sylow subgroup of $\bar{G}$ and using the structure of inertia gives the required finiteness. Case (ii) follows similarly from CG II, Proposition 12 after completion.

Theorem 2.3. Let A be a strictly local ring, let k be its residue field, and let K be the total ring of fractions of A. If K is noetherian and if $\ell \neq \operatorname{char} k$, then

$$\operatorname{cd}_\ell K \leq \dim A.$$

The proof proceeds by noetherian induction and reduction to the case of a domain. One uses normalization, passage to finite extensions, and Theorem 2.2 applied to henselian localizations at codimension-one points. This is the same dévissage as in the classical proof for excellent local rings; the strict locality ensures that no residue-field contribution remains. For example, a strictly henselian discrete valuation ring has fraction field of ℓ-cohomological dimension at most one for ℓ prime to the residue characteristic.

Corollary 2.4. Let A be a strictly local noetherian ring, let K be its total ring of fractions, and let ℓ be invertible in A. Then $\operatorname{cd}_\ell K \leq \dim A$.

Corollary 2.5. Let X be a noetherian scheme, let x be a point, and let $X_{\bar{x}}$ be a strict localization at a geometric point over x. If ℓ is invertible at x, the total ring of fractions of $X_{\bar{x}}$ has ℓ-cohomological dimension at most $\operatorname{codim} x$.

3. Fraction fields of a strictly local ring

3.0

Let A be strictly local with closed point s, and let $U = \operatorname{Spec} A - \{s\}$. The preceding estimates for fraction fields will be applied to the fibres of direct images from U to $\operatorname{Spec} A$. The basic point is that the strict localization at a geometric point sees only the corresponding henselian local algebra, so that the cohomology of the punctured local scheme is controlled by the cohomological dimension of its total fraction ring.

Proposition 3.2. Let A be strictly local noetherian of dimension n, and let ℓ be invertible in A. Then for every ℓ-torsion sheaf F on $U = \operatorname{Spec} A - \{s\}$, one has

$$H^q(U, F) = 0 \qquad (q > 2n - 1).$$

If F is the restriction of a sheaf from $\operatorname{Spec} A$, the bound improves in the expected low-dimensional cases.

Proof. The proof uses the Leray spectral sequence for the immersion $j : U \to \operatorname{Spec} A$. Its higher direct images are computed on strict localizations, and Corollary 2.5 gives the necessary vanishing of their fibres. Induction on the dimension of the support then gives the bound.

Lemma 3.3. Let A be strictly local noetherian and let $j : U \to \operatorname{Spec} A$ be the complement of the closed point. If ℓ is invertible in A, then the sheaves $R^q j_* F$ have support constrained by codimension: their stalks vanish at a point once q exceeds the codimension of that point in the relevant strict localization.

This is just the stalkwise form of Corollary 2.5 and is the technical input for the estimates in the next section.

4. Cohomological dimension when ℓ is invertible in $\mathcal{O}_X$

Theorem 4.1. Let X be a noetherian scheme of dimension n, and let ℓ be invertible on X. Suppose that the ℓ-cohomological dimension of every residue field of X is bounded by N. Then

$$\operatorname{cd}_\ell X \leq N + 2n.$$

More precisely, if an ℓ-torsion sheaf F is supported in dimension $\leq d$, then its cohomology vanishes above the corresponding bound $N + 2d$.

Proof. It is enough, by the constructible filtration theorem of IX, to treat constructible sheaves, then to proceed by induction on the support. On a dense open subset of each irreducible component the sheaf becomes locally constant and isotrivial; after a finite *étale* cover and using the trace method of IX, 5, one reduces to constant sheaves. The local terms are controlled by Proposition 3.2 and Lemma 3.3. The spectral sequence for the inclusion of the generic points gives the bound.

Corollary 4.2. Let X be of finite type over a field k, with $\ell \neq \operatorname{char} k$. Then

$$\operatorname{cd}_\ell X \leq \operatorname{cd}_\ell k + 2 \dim X.$$

In particular, if k is separably closed, $\operatorname{cd}_\ell X \leq 2 \dim X$.

Corollary 4.3. Let X be of finite type over a separably closed field, and let ℓ be invertible on X. Then $H^q(X, F) = 0$ for every ℓ-torsion sheaf F and every $q > 2 \dim X$.

Corollary 4.4. Let X be an algebraic curve over a separably closed field, with $\ell \neq \operatorname{char} k$. Then $\operatorname{cd}_\ell X \leq 2$; if X is affine, the sharper bound $\operatorname{cd}_\ell X \leq 1$ is supplied later by XIV.

Example 4.5. One cannot hope for an analogous local formula in terms only of the local rings of X. There are discrete valuation rings with algebraically closed residue field whose fraction fields have arbitrarily large cohomological dimension; they arise by taking formal curve arcs through a rational point of a smooth variety whose function field is prescribed.

4.6

These estimates will be used repeatedly in the sequel. Their force is that, for ℓ invertible, the relevant cohomological dimension is twice the geometric dimension, plus the cohomological dimension of the ground field.

5. Cohomological dimension in the case $\ell = p$

5.0

Using Artin–Schreier theory (IX, 3.5), one obtains a better bound for the p-cohomological dimension of a scheme of characteristic p. Let $\operatorname{cd}_{\mathrm{qc}} X$ be the largest integer n such that $H^n(X, F) \neq 0$ for some quasi-coherent $\mathcal{O}_X$-module F in the Zariski sense.

Theorem 5.1. Let X be a noetherian scheme of characteristic $p > 0$. Then

$$\operatorname{cd}_p X \leq \operatorname{cd}_{\mathrm{qc}} X + 1.$$

In particular, a noetherian affine scheme of characteristic $p > 0$ has p-cohomological dimension at most one.

Proof. Apply IX, 5.5. Since higher direct images under finite morphisms vanish for abelian sheaves, and likewise for quasi-coherent modules, one reduces to proving $H^q(X, i_!(\mathbb{Z}/p\mathbb{Z})_U) = 0$ for an open immersion $i : U \to X$ and $q > \operatorname{cd}_{\mathrm{qc}} X + 1$. Let Y be a closed subscheme with support $X - U$, defined by a coherent ideal J. The Artin–Schreier map $f \mapsto f^p - f$ induces a morphism $J \to J$, and the Artin–Schreier sequence gives

$$0 \longrightarrow i_!(\mathbb{Z}/p\mathbb{Z})_U \longrightarrow J \longrightarrow J \longrightarrow 0.$$

Since J is coherent, its cohomology vanishes above $\operatorname{cd}_{\mathrm{qc}} X$, giving the assertion.

Corollary 5.2. Let X be of finite type over a field k of characteristic $p > 0$. Then

$$\operatorname{cd}_p X \leq \dim X + 1.$$

If k is separably closed and X is quasi-projective, then

$$\operatorname{cd}_p X \leq \dim X.$$

The first assertion is clear from $\operatorname{cd}_{\mathrm{qc}} X \leq \dim X$. For the second, the top coherent cohomology groups are finite-dimensional k-vector spaces. In the exact Artin–Schreier sequence above, the induced endomorphism of the top cohomology is of the form $\epsilon - \mathrm{id}$, with $\epsilon(rv) = r^p \epsilon(v)$. The corresponding additive morphism of affine space has Jacobian $-I$, hence is *étale*; over a separably closed field it is surjective. The top obstruction vanishes.

6. Cohomological dimension for schemes of finite type over $\operatorname{Spec}\mathbb{Z}$

Proposition 6.1. Let X be quasi-finite over $\operatorname{Spec}\mathbb{Z}$. Suppose either $\ell \neq 2$, or every residue field of X of characteristic zero is totally imaginary. Then $\operatorname{cd}_\ell X \leq 3$.

This follows from Theorem 2.2(ii), Corollary 4.2, and the fact that a number field, totally imaginary when $\ell = 2$, has ℓ-cohomological dimension at most two (CG II, 4.4). Class field theory shows that equality holds if $X \to \operatorname{Spec}\mathbb{Z}$ is proper and surjective.

The results of Section 4 do not apply unchanged when X is merely of finite type over $\operatorname{Spec}\mathbb{Z}$, because the hypotheses at points of residual characteristic ℓ fail in general. The estimate of Theorem 5.1 must be used in that characteristic.

Theorem 6.2. Let X be of finite type over $\operatorname{Spec}\mathbb{Z}$, and let ℓ be a prime. If $\ell = 2$, suppose that no residue field of X is orderable. Then

$$\operatorname{cd}_\ell X \leq 2\dim X + 1.$$

More precisely, let F be an ℓ-torsion sheaf such that, for every point x with $F_{\bar{x}} \neq 0$,

$$\dim \overline{\{x\}} \leq \begin{cases} (N-1)/2, & \ell \neq \operatorname{char} k(x), \\ N-1, & \ell = \operatorname{char} k(x). \end{cases}$$

Then $H^p(X, F) = 0$ for $p > N$.

Proof. Proceed by induction on X, and reduce to constructible F by writing any sheaf as the filtered limit of its constructible subsheaves. The set of points where $F_{\bar{x}} \neq 0$ is then constructible, and induction handles the case where it is not dense. If the support is in characteristic ℓ, Theorem 5.1 gives the bound; if it is in characteristic $p \neq \ell$, Corollary 4.3 and the fact $\operatorname{cd}_\ell \mathbb{F}_p = 1$ give it.

For a reduced X, write $X = X_0 \cup X_1$, where X_1 is the union of the irreducible components of characteristic ℓ, and X_0 the union of the others. The exact sequence

$$0 \to F \to j_{0*}j_0^*F \oplus j_{1*}j_1^*F \to C \to 0$$

has C supported on $X_0 \cap X_1$, which is of characteristic ℓ and dimension at most $\dim X_1 - 1$. The induction and Theorem 5.1 reduce the assertion to the cases $X = X_0$ and $X = X_1$; the latter was already handled.

It remains to treat the case in which every maximal point has characteristic different from ℓ. Let $i : x \to X$ be the generic point of an irreducible component. Using the exact sequence comparing F with the sum of its generic-point extensions, induction reduces to $F = i_* G$. The spectral sequence

$$H^p(X, R^q i_* G) \Longrightarrow H^{p+q}(\operatorname{Spec} k(x), G)$$

finishes the proof. If $s = \dim X$, then $k(x)[i]$ has transcendence degree $s-1$ over $\mathbb{Q}[i]$, a field of cohomological dimension two; Theorem 2.1 gives $\operatorname{cd}_\ell k(x) = s+1$, under the non-orderability hypothesis when $\ell = 2$. The stalks of $R^q i_* G$ vanish when q exceeds the appropriate codimension, by Proposition 3.2 and Theorem 2.2. Applying induction to these higher direct images gives the required vanishing.

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SGA 4, Exposé XI

Comparison with Classical Cohomology: the Smooth Case

M. Artin

1. Introduction

In Section 4 we shall prove that *étale* cohomology with values in a locally constant torsion sheaf agrees with classical cohomology for a scheme X smooth over $\operatorname{Spec}\mathbb{C}$. The proof, essentially the one given in Giraud's Bourbaki exposé, is elementary once one uses Grauert's theorem recalled below (4.3(iii)), namely that every finite covering of $X(\mathbb{C})$ comes from an *étale* covering of X. We give it here even though a more complete result will be proved later using Hironaka's resolution of singularities. The good neighbourhoods constructed in Section 3 may also be useful in other situations.

It is not true that *étale* cohomology and classical cohomology agree for non-torsion sheaves. For example, $H^1(X_{\text{ét}}, \mathbb{Z}) = 0$ for every normal scheme X (IX, 3.6(ii)), whereas the corresponding classical cohomology need not vanish.

2. Existence of sufficiently general hyperplane sections

We shall use the following classical result, for which the original notes lacked a convenient reference.

Theorem 2.1. Let k be a field, and let $V \subset \mathbf{P}^n_k$ be a projective algebraic scheme purely of dimension r. Let V' be the open subscheme of points of V smooth over $\operatorname{Spec} k$, and let P be a rational point of $\mathbf{P}^n$. Then:

(i) A sufficiently general linear subspace L of prescribed dimension d in $\mathbf{P}^n$ meets V transversally at every point of $L \cap V'$.

(ii) Let $\{H_1, \ldots, H_s\}$, $s \leq r$, be hypersurfaces of degree $d \geq 2$ through P, and put $Y = H_1 \cap \cdots \cap H_s$. If $\{H_1, \ldots, H_s\}$ is sufficiently general, then Y meets V transversally at every point of $Y \cap V'$.

Here "sufficiently general" means that the relevant parameter space, a Grassmannian in (i) or a product of projective spaces in (ii), contains a dense open subset over which the assertion holds. Thus it does not assert the existence of a rationally defined member when the ground field is finite.

Proof. For (i), induction reduces immediately to the case of a hyperplane. After covering V' by finitely many opens, one may assume an affine situation $V \subset \mathbb{A}^n_k = \operatorname{Spec} k[x_1, \ldots, x_n]$. The hyperplane has equation

$$\ell(x) = a_0 + \sum_{i=1}^{n} a_i x_i.$$

Since a smooth scheme is locally a complete intersection, after shrinking one may suppose that V is cut out near V' by equations $f_1, \ldots, f_{n-r}$ with an invertible Jacobian minor. The condition that L fail to meet V transversally is then expressed by the existence of a point $x \in V'$ satisfying an explicit finite system of equations in the parameters a_i. The image of this closed condition is constructible by EGA IV, 9.5.1. It is enough to check the generic parameter, where the equations plainly have no solution. This proves (i).

For (ii), use the rational map $\mathbf{P}^n \dashrightarrow \mathbf{P}^N$ defined by the linear system of degree d hypersurfaces through P. For $d \geq 2$ it is the blow-up of P followed by a locally closed immersion off P. Away from P the assertion reduces to (i) applied to the image. At P the conclusion is immediate for sufficiently general hypersurfaces.

3. Construction of good neighbourhoods

Definition 3.1. An *elementary fibration* is a morphism of schemes $f : X \to S$ which can be embedded in a commutative diagram

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[dr, "f"'] & \bar{X} \arrow[d, "\bar{f}"] & Y \arrow[l, hook', "i"'] \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

satisfying:

(i) j is an open immersion, dense in every fibre, and $X = \bar{X} - Y$;

(ii) $\bar{f}$ is smooth and projective, with geometrically irreducible fibres of dimension one;

(iii) g is an *étale* covering, and every fibre of g is non-empty.

Such an embedding of X, if it exists, is easily checked to be unique up to canonical isomorphism.

Definition 3.2. A *good neighbourhood relative to* S is an S-scheme X for which there exist S-schemes

$$X = X_n, \dots, X_0 = S$$

and elementary fibrations $f_i : X_i \to X_{i-1}$, $i = 1, \dots, n$.

Proposition 3.3. Let k be algebraically closed, let $X/\operatorname{Spec} k$ be smooth, and let $x \in X$ be a rational point. Then there is an open neighbourhood of x in X which is a good neighbourhood relative to $\operatorname{Spec} k$.

Proof. Assume X irreducible. By induction on $n = \dim X$, it is enough to find a neighbourhood U of x and an elementary fibration $f : U \to V$, with V smooth of dimension $n-1$. Indeed, a good neighbourhood of $f(x)$ in V then pulls back to one of x in U.

Embed X as an affine open in $\mathbb{A}^r$, let X_0 be its closure in $\mathbf{P}^r$, let $\bar{X}$ be the normalization of X_0, and let $Y = \bar{X} - X$ with reduced structure. Let $S \subset \bar{X}$ be the singular locus; then $S \subset Y$, $\dim \bar{X} = n$, $\dim Y = n-1$, and $\dim S \leq n-2$.

Choose a projective embedding of $\bar{X}$ by an invertible sheaf which is a sufficiently high power of a very ample sheaf. By Theorem 2.1 there exist hyperplanes $H_1, \dots, H_{n-1}$ through x such that $L = H_1 \cap \dots \cap H_{n-1}$ meets $\bar{X}$ and Y transversally. Then $\bar{X} \cap L$ is a smooth connected curve, and $Y \cap L$ is zero-dimensional. Choose an additional hyperplane H_0 such that H_0 meets $\bar{X} \cap L$ transversally and avoids $Y \cap L$.

Projection from the center $C = H_0 \cap \cdots \cap H_{n-1}$ gives a rational map $\mathbf{P}^N \dashrightarrow \mathbf{P}^{n-1}$. Blow up C, take the proper transform $\bar{X}'$, put $X' = X - \bar{X} \cap C$, and let $Y' = \bar{X}' - X'$. We get a diagram

$$\begin{tikzcd}
X' \arrow[r, "j"] \arrow[dr, "f'"'] & \bar{X}' \arrow[d, "\bar{f}'"] & Y' \arrow[l, "i"'] \arrow[dl, "g'"] \\
& \mathbf{P}^{n-1} &
\end{tikzcd}$$

Near the point $v = f'(x)$, the fibre of $\bar{f}'$ is identified with the smooth curve $\bar{X} \cap L$; by Hironaka's lemma, smoothness near that fibre follows from smoothness at its generic point. The morphism g' is *étale* near v because L meets Y transversally, and each fibre is non-empty. After replacing $\mathbf{P}^{n-1}$ by a neighbourhood of v, the diagram gives an elementary fibration.

4. The comparison theorem

4.0

Let X be a scheme locally of finite type over $\operatorname{Spec}\mathbb{C}$, and put on $X(\mathbb{C})$ its usual topology. Denote by X_{cl} the site whose objects are local isomorphisms $U \to X(\mathbb{C})$, that is, maps of topological spaces which are locally homeomorphisms onto open subsets of $X(\mathbb{C})$. A family $\{U_\nu \to U\}$ is covering if the images cover U. Since open immersions are local isomorphisms, the site X_{cl} has the same associated topos as the usual topological space $X(\mathbb{C})$. Thus one may compute ordinary cohomology using X_{cl}.

If $f : X' \to X$ is *étale*, the analytic map $f(\mathbb{C}) : X'(\mathbb{C}) \to X(\mathbb{C})$ is a local isomorphism, by the Jacobian criterion and the implicit function theorem. Therefore the functor $X' \mapsto X'(\mathbb{C})$ defines a morphism of sites

$$\epsilon : X_{\mathrm{cl}} \longrightarrow X_{\text{ét}}. \tag{4.2}$$

Theorem 4.3.

(i) X is connected and non-empty if and only if $X(\mathbb{C})$ is connected and non-empty.

(ii) The functor ϵ is fully faithful.

(iii) (Grauert–Remmert theorem.) The functor ϵ induces an equivalence between finite coverings of $X(\mathbb{C})$, with its usual locally compact topology, and *étale* coverings of X.

We accept (i) as known; it follows, for example, from GAGA. Assertion (ii) follows easily from (i): for a morphism $\varphi : X'(\mathbb{C}) \to X''(\mathbb{C})$ over $X(\mathbb{C})$, after reducing to the connected affine separated case, its graph is a connected component of $(X' \times_X X'')(\mathbb{C})$, hence comes from a connected component of $X' \times_X X''$. This component is the graph of a morphism $X' \to X''$. For (iii), by descent one reduces to X connected normal and affine. Taking a finite morphism $X \to \mathbb{A}^n$, any finite connected covering of $X(\mathbb{C})$ becomes an analytic covering of $\mathbb{A}^n(\mathbb{C})$, extends to an analytic covering of $\mathbf{P}^n(\mathbb{C})$ by Grauert–Remmert, and the resulting coherent analytic algebra is algebraic by GAGA. Taking $Y = \operatorname{Spec} A$ gives the desired *étale* covering.

Theorem 4.4. Let X be smooth over $\operatorname{Spec}\mathbb{C}$.

(i) The category of constructible locally constant torsion sheaves on $X_{\text{ét}}$ is equivalent to the category of locally constant torsion sheaves with finite fibres on X_{cl}; the quasi-inverse functors are ϵ_* and ϵ^*.

(ii) If F is a locally constant torsion sheaf with finite fibres on X_{cl}, and if the induced sheaf on $X_{\text{ét}}$ is again denoted F, then
$$R^q\epsilon_*F = 0 \quad (q > 0).$$

(iii) With the notation of (i), the canonical map
$$H^q(X_{\text{ét}}, F) \longrightarrow H^q(X_{\text{cl}}, F)$$
is bijective for every q. In particular,
$$H^q(X_{\text{ét}}, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} H^q(X_{\text{cl}}, \mathbb{Z}/n\mathbb{Z})$$
for every q.

Proof. For (i), the sheaves in question are precisely those represented by finite *étale* coverings of X, respectively finite coverings of $X(\mathbb{C})$. The equivalence of coverings is Theorem 4.3(iii). Assertion (iii) follows from (ii) and the Leray spectral sequence for ϵ. It remains to prove (ii).

The sheaf $R^q\epsilon_*F$ is associated with the presheaf $X' \mapsto H^q(X'_{\text{cl}}, F)$ on $X_{\text{ét}}$. We must therefore show that every cohomology class becomes zero after passing to an *étale* neighbourhood of any prescribed point.

Lemma 4.5. Let $\xi \in H^q(X_{\text{cl}}, F)$, and let $x \in X(\mathbb{C})$. There exists an *étale* morphism $X' \to X$, whose image contains x, such that the image of ξ in $H^q(X'_{\text{cl}}, F)$ is zero.

Proof. The proof is by induction on $n = \dim X$. The problem is local for the *étale* topology, so one may assume F constant, and by dévissage even $F = \mathbb{Z}/n\mathbb{Z}$. After replacing X by a Zariski neighbourhood of x, Proposition 3.3 allows us to assume that X admits an elementary fibration $f : X \to S$. With the notation of Definition 3.1, a direct calculation gives $R^q j_{\text{cl}*}F = 0$ for $q > 1$, the sheaf $R^1 j_{\text{cl}*}F$ is extension by zero of a constant sheaf on Y_{cl}, and $j_{\text{cl}*}F$ is the constant sheaf $\mathbb{Z}/n\mathbb{Z}$ on $\bar{X}$. Moreover $R^q \bar{f}_{\text{cl}*}(j_{\text{cl}*}F)$ is computed fibrewise. The Leray spectral sequence implies that
$$R^0 f_{\text{cl}*}\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/n\mathbb{Z}, \quad R^1 f_{\text{cl}*}\mathbb{Z}/n\mathbb{Z} \text{ is locally constant torsion}, \quad R^q f_{\text{cl}*}\mathbb{Z}/n\mathbb{Z} = 0 \ (q > 1).$$
Thus the Leray sequence reduces to
$$\cdots \to H^q(S_{\text{cl}}, f_{\text{cl}*}F) \to H^q(X_{\text{cl}}, F) \to H^{q-1}(S_{\text{cl}}, R^1 f_{\text{cl}*}F) \to \cdots . \tag{*}$$
Let $s = f(x)$. By the induction hypothesis and (i), every class on S with finite locally constant torsion coefficients dies after an *étale* neighbourhood $S' \to S$ of s. Since the functors $R^q f_{\text{cl}*}$ commute with *étale* base change, the pullback $X \times_S S' \to X$ is an *étale* neighbourhood of x, and the exact sequence (*) kills first the boundary term and then the remaining term. The case $q = 1$ also follows directly from the classification of principal finite coverings and Theorem 4.3(iii).

Variant 4.6. Serre gave a more elegant proof of Lemma 4.5. A connected good neighbourhood X satisfies:

(i) $\pi_n(X(\mathbb{C})) = 0$ for $n > 1$;

(ii) $\pi_1(X(\mathbb{C}))$ is an iterated extension of finitely generated free groups.

Indeed, for an elementary fibration $X \to S$, the map $X(\mathbb{C}) \to S(\mathbb{C})$ is a locally trivial fibration whose fibre is a non-complete smooth connected curve; such a fibre has vanishing higher homotopy and free fundamental group. The long exact homotopy sequence gives the assertions by induction. Hence $X(\mathbb{C})$ is a $K(\pi, 1)$. A cohomology class becomes zero on the universal cover; it remains to show that it becomes zero on a finite cover. This is equivalent to the group π_1 being good in Serre's sense, i.e.

$$H^q(\widehat{\pi}_1, M) \xrightarrow{\sim} H^q(\pi_1, M)$$

for every finite continuous π_1-module M. Iterated extensions of finitely generated free groups are good (CG I, pp. 15–16, Exercises 1, 2).

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SGA 4, Exposé XII

The Proper Base Change Theorem

M. Artin

1. Introduction

In this exposé we give the fundamental theorem which says, in a slightly more precise form, that for a *proper* morphism $f : X \to Y$ of schemes and for an abelian torsion sheaf F on X, the fibre of $R^q f_* F$ at a geometric point ξ of Y is isomorphic to the cohomology

$$H^q(X_\xi, F|_{X_\xi})$$

of the fibre X_ξ.

For a proper morphism of paracompact spaces, the analogous result is well known and easy (Godement II, 4.11). For schemes it is much subtler. In particular, the torsion hypothesis on F is really necessary, unlike in the paracompact topological case.

The proof proceeds in several stages. By more or less formal dévissage one reduces to the case where Y is noetherian, the relative dimension of f is at most one, and F is constant. To treat that case one uses the specialization theorem for the fundamental group (5.9), a non-abelian variant of the theorem stated above; this specialization theorem can be proved directly in the special case required here, as is done in the next exposé.

2. An example

Let us show that the torsion hypothesis on F is necessary. Let X be a nonsingular surface over an algebraically closed field k, let Y be a nonsingular curve over k, and let $f : X \to Y$ be a proper morphism whose generic fibre is smooth and whose geometric fibre at a geometric point $\bar{y}$ of Y is irreducible and has one ordinary double point. Let F be the constant sheaf $\mathbb{Z}_X$.

Recall the following fact: if z is any point of a scheme Z and $i : z \to Z$ is the inclusion, then $H^1(Z, i_*\mathbb{Z}_z) = 0$ (IX, 3.6(i)). Since X is normal, hence unibranch, $\mathbb{Z}_X \simeq i_*\mathbb{Z}_x$, where $i : x \to X$ is the inclusion of the generic point. Thus $H^1(X, \mathbb{Z}_X) = 0$, and likewise $R^1 f_* \mathbb{Z}_X = 0$. But for the special geometric fibre X_0, with its ordinary double point, one has

$$H^1(X_0, \mathbb{Z}_{X_0}) \simeq \mathbb{Z}.$$

Indeed, if $i_0 : x_0 \to X_0$ is the inclusion of the generic point and Q is the singular point, the curve has two branches at Q. Hence the fibre of $i_{0*}\mathbb{Z}_{x_0}$ at Q is $\mathbb{Z} \times \mathbb{Z}$, while away from Q the curve is normal and $i_{0*}\mathbb{Z}_{x_0} \simeq \mathbb{Z}_{X_0}$. There is an exact sequence

$$0 \longrightarrow \mathbb{Z}_{X_0} \longrightarrow i_{0*}\mathbb{Z}_{x_0} \longrightarrow (\mathbb{Z})_Q \longrightarrow 0,$$

where the last term is extension by zero from Q. All three terms have $H^0 \simeq \mathbb{Z}$, and $H^1(X_0, i_{0*}\mathbb{Z}_{x_0}) = 0$; the long exact sequence gives the displayed isomorphism.

3. Reminders on non-abelian H^1

We briefly recall the non-abelian cohomological facts needed below. Most proofs are omitted; they are well known in various special cases and can be found in Giraud's thesis.

Let X be a scheme and F a sheaf of groups on X. For every *étale* X'/X, there is a pointed set $H^1(X', F)$, covariantly functorial in F and contravariantly functorial in X'. It commutes with finite products of sheaves. If $f : X \to Y$ is a morphism, define a sheaf of pointed sets $R^1 f_* F$ as the sheaf associated with the presheaf

$$Y' \longmapsto H^1(X \times_Y Y', F),$$

for $Y' \to Y$ *étale*.

Proposition 3.1. Let $0 \to F \xrightarrow{u} G$ be an injection of sheaves of groups, and let $C = G/F$ be the sheaf of homogeneous spaces under G. There is an exact sequence of sheaves of pointed sets

$$0 \to f_*F \to f_*G \to f_*C \xrightarrow{\partial} R^1 f_*F \xrightarrow{u^1} R^1 f_*G.$$

This is obtained by sheafifying, for every *étale* Y'/Y, the usual exact sequence

$$0 \to F(X') \to G(X') \to C(X') \xrightarrow{\partial} H^1(X', F) \to H^1(X', G),$$

where $X' = X \times_Y Y'$. The equivalence relation induced by ∂ is the one arising from the action of f_*G on f_*C; the equivalence relation induced by u^1 is described in the usual way by twisting.

Proposition 3.2. For morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$, and every $Z' \to Z$ *étale*, putting $Y' = Y \times_Z Z'$ and $X' = X \times_Z Z'$, there is an exact sequence

$$0 \to H^1(Y', f_*F) \to H^1(X', F) \to H^0(Y', R^1 f_*F).$$

After sheafification this gives an exact sequence of pointed sheaves on Z:

$$0 \to (R^1 g_*) f_*F \to R^1(gf)_*F \to g_*(R^1 f_*F).$$

Proposition 3.3. The functors $H^1(X', F)$, for X' *étale* over X, and $R^1 f_*F$, for arbitrary $f : X \to Y$, are effaceable. More precisely, there exists an injection $F \to G$ such that all pointed sets $H^1(X', G)$ and all sheaves $R^1 f_*G$ are zero. If X is noetherian and F is a constructible sheaf of ind-finite groups, one may choose G again a sheaf of ind-finite groups.

Proof. For the *étale* topology the standard Godement resolution suffices:

$$F \longrightarrow G = \prod_{x \in X} i_{x*} i_x^* F, \tag{3.4}$$

where x runs through the points of X and $i_x : \bar{x} \to X$ is a geometric point over x. For X'/X *étale*, the sections of G are products over the geometric points above those of X, and the map $F(X') \to G(X')$ is injective. Since over a geometric point every *étale* scheme is a disjoint sum of spectra of separably closed fields, the H^1 of each summand vanishes; Proposition 3.2 then gives $H^1(X', G) = 0$, and hence $R^1 f_*G = 0$ for every f. If F is constructible on noetherian X, finitely many geometric points already suffice for injectivity, and the finite product remains a sheaf of ind-finite groups by IX, 1.4 and IX, 1.6.

3.5

As in abelian cohomology (V, 2.4), non-abelian H^1 may be computed by the Čech procedure. We shall use this without further comment.

4. The base-change morphism

Let

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y \end{tikzcd}$$

be a cartesian square of schemes, and let F be a sheaf on X. There is a canonical base-change morphism

$$g^* R^q f_* F \longrightarrow R^q f'_* g'^* F. \tag{4.1}$$

For $q = 0$ it is induced directly by adjunction. For general q, choose an injective or effacing resolution and use the compatibility of pullback with the construction of direct images; in the non-abelian case $q = 1$ the same construction gives a morphism of pointed sheaves

$$g^* R^1 f_* F \longrightarrow R^1 f'_* g'^* F. \tag{4.2}$$

When Y' is the spectrum of a separably closed field over a geometric point ξ of Y, this morphism is the canonical map from the fibre of $R^q f_* F$ at ξ to the cohomology of the geometric fibre X_ξ.

Remark 4.3. The construction is functorial in the square and compatible with composition of base changes. It is also compatible with the exact sequences of Section 3 and with the usual long exact sequences in abelian cohomology.

Proposition 4.4. Suppose that the base-change morphism is an isomorphism for a class of sheaves stable under extensions and under the operations used in the exact sequences above. Then it remains an isomorphism after dévissage through these exact sequences. In particular, for torsion abelian sheaves it is enough to verify proper base change after reducing to constructible sheaves and then to constant sheaves on suitable finite covers.

Proof. For abelian sheaves this is the usual five-lemma argument in the long exact sequence of cohomology associated with a short exact sequence. For H^1 of groups one uses the exact sequence of pointed sets from Proposition 3.1. The compatibility of the boundary maps with base change is built into the construction of (4.1) and (4.2). Thus the property is preserved under kernels, cokernels when meaningful, extensions, and the reductions supplied by IX, 5.5.

5. Statement of the main theorem and some variants

Theorem 5.1. Let $f : X \to Y$ be a proper morphism of schemes, and let F be an abelian torsion sheaf on X. For every morphism $g : Y' \to Y$, form the cartesian square

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y. \end{tikzcd}$$

Then for every $q \geq 0$ the base-change morphism

$$g^* R^q f_* F \xrightarrow{\sim} R^q f'_* g'^* F$$

is an isomorphism. Equivalently, for every geometric point ξ of Y,

$$(R^q f_* F)_\xi \simeq H^q(X_\xi, F|_{X_\xi}).$$

The theorem is the proper base change theorem. It is first stated here with torsion coefficients; the torsion condition cannot be omitted, as the example of Section 2 shows.

Corollary 5.2. Under the hypotheses of Theorem 5.1, if F is constructible, then the formation of $R^q f_* F$ commutes with every base change. In particular, its fibres are computed on the corresponding geometric fibres.

Corollary 5.3. Let S be the spectrum of a henselian local ring, let s_0 be its closed point, and let $f : X \to S$ be proper. If X_0 is the closed fibre and F is a torsion sheaf on X, then

$$H^q(X, F) \xrightarrow{\sim} H^q(X_0, F|_{X_0})$$

for all q, whenever the relevant support and constructibility hypotheses reduce the assertion to Theorem 5.1.

Corollary 5.3 bis. The same assertion holds after replacing S by its strict henselization at a geometric point; this is simply the fibre formulation of Theorem 5.1.

Corollary 5.4. Let $f : X \to S$ be proper, with S henselian local and closed fibre X_0. For torsion sheaves satisfying the preceding hypotheses, the restriction morphism on cohomology from X to X_0 is an isomorphism. The same remains true for suitable non-abelian H^1 statements with finite or ind-finite structure groups.

Corollary 5.5. Let S be the spectrum of a henselian local ring, with closed point s_0, and let $f : X \to S$ be proper. Put $X_0 = X \times_S s_0$. Then:

(i) For a finite constant sheaf of sets F, the map $F(X) \to F(X_0)$ is bijective.

(ii) For a finite constant sheaf of groups G, the map

$$H^1(X, G_X) \longrightarrow H^1(X_0, G_{X_0})$$

is bijective.

(iii) For an abelian torsion sheaf F, the restriction map

$$H^q(X, F) \longrightarrow H^q(X_0, F|_{X_0})$$

is an isomorphism in the cases covered by Theorem 5.1.

For constant sheaves, (i) is the assertion that connected components are preserved by passage to the closed fibre. For finite constant groups, (ii) is the corresponding assertion for principal finite Galois coverings.

5.6

Let $F = T_X$ be a constant sheaf of sets on X, with value T. Then $F(X)$ is the set of locally constant maps from X to T. If $X_0 \subset X$ is a subscheme and T has at least two elements, the map $F(X) \to F(X_0)$ is bijective exactly when the map from clopen subsets of X to clopen subsets of X_0 is bijective. If X_0 has finitely many connected components, this is equivalent to bijectivity of $\Pi_0(X_0) \to \Pi_0(X)$. Thus Corollary 5.5(i) may be restated as follows.

Corollary 5.7. Let S be the spectrum of a henselian ring, with closed point s_0. Let $f : X \to S$ be proper and let X_0 be the closed fibre. Then the map

$$\Pi_0(X_0) \longrightarrow \Pi_0(X)$$

induced by $X_0 \to X$ is bijective.

In particular $\Pi_0(X)$ is finite. In the noetherian case this can be proved directly.

Lemma 5.8. Corollary 5.7 holds when S is noetherian.

Proof. Since X is noetherian, $\Pi_0(X)$ is finite. Replacing X by a connected component, assume X is connected and non-empty. The image of X in S is closed and non-empty, hence contains the closed point, so $X_0 \neq \varnothing$. Let

$$X \xrightarrow{f'} S' \xrightarrow{\pi} S$$

be the Stein factorization of f. Then S'/S is finite and S' is connected, while the fibres of f' are connected and non-empty. The space X_0 is the union of the fibres over the closed points of S'. It remains to show that S' is the spectrum of a local ring. Since S is henselian, the finite S-algebra defining S' is a product of local rings (VII, 4.1(i)); because S' is connected, only one factor occurs.

5.9.0

If $F = G_X$ is a constant sheaf of groups with finite value G, then $H^1(X, G_X)$ classifies principal Galois coverings of X with group G. Hence Corollary 5.5(ii), together with (i), is equivalent to the following specialization theorem for the fundamental group.

Theorem 5.9. With the notation of Corollary 5.5, suppose X is connected and non-empty. Then X_0 is connected and non-empty, and for every geometric point $\bar{x}_0$ of X_0, the morphism of fundamental groups

$$\Pi_1(X_0, \bar{x}_0) \longrightarrow \Pi_1(X, \bar{x}_0)$$

is an isomorphism.

Equivalently, by Galois theory for coverings:

Theorem 5.9 bis. Let S be the spectrum of a henselian ring, with closed point s_0, let $f : X \to S$ be proper, and let X_0 be the closed fibre. For a scheme Z, denote by $\mathrm{Et}(Z)$ the category of *étale* coverings of Z. Then the restriction functor

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_0)$$

is an equivalence of categories.

Remark 5.10. Theorem 5.9 was already proved in SGA 1, X, 2.1 when S is the spectrum of a complete noetherian local ring. The proof given here and in the following exposé is of a quite different nature.

5.11

One may also formulate a non-commutative variant of Corollary 5.5(ii) using Giraud's two-dimensional cohomology. Assume S noetherian. Let L be a link on X, locally represented in the *étale* topology by a sheaf of ind-finite groups. Consider

$$\varphi^{(2)} : H^2(X, L) \longrightarrow H^2(X_0, L_0),$$

where L_0 is the restriction of L to X_0. Then

$$(\varphi^{(2)})^{-1}(H^2(X_0, L_0)') = H^2(X, L)', \tag{5.11.1}$$

where primes denote the subsets of neutral elements.

Equivalently, for every gerbe $\mathcal{G}$ on X with link L as above, if $\mathcal{G}_0$ denotes its restriction to X_0, the restriction functor on categories of sections

$$\mathcal{G}(X) \longrightarrow \mathcal{G}_0(X_0)$$

is an equivalence. Full faithfulness follows from Corollary 5.5(i), and essential surjectivity when $\mathcal{G}(X) \neq \varnothing$ follows from Corollary 5.5(ii). The remaining assertion is precisely that neutrality after restriction to X_0 implies neutrality over X, which is (5.11.1). The proof of (5.11.1) is not given here; one can show more generally, for noetherian X and a closed subscheme, that validity of Corollary 5.5(i) and (ii) implies (5.11.1), using the descent result VIII, 9.4(a).